

Antisymmetric breathing in altermagnetic skyrmions

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A skyrmion in an altermagnet with d wave symmetry consists of two elliptical sublattice textures with perpendicular long axes. For each sublattice component $\eta = A, B$, we define an effective skyrmion radius $R_\eta = \sqrt{a_\eta b_\eta}$, where a_η and b_η are the distances from the skyrmion center to its boundary along y and x , respectively. For the anisotropic exchange parameters studied, the equilibrium radius at zero field is smaller than in a reference with parallel sublattice textures and otherwise identical parameters. A magnetic field perpendicular to the film expands one sublattice skyrmion and contracts the other, generating a radius difference $\delta R = R_A - R_B$ and a net magnetic moment. The exchange modulation used to represent uniaxial strain also produces a nonzero radius difference at zero field. Because the strong and weak exchange directions are interchanged between the two sublattices, a common directional change of the exchange couplings increases one radius and decreases the other. Unlike the magnetic coupling, this mechanism vanishes when the two sublattice exchange tensors become identical. The radius difference also supports an antisymmetric breathing mode. After a short field pulse, it oscillates in quadrature with the uniform helicity, the common rotation of the wall magnetization within the film plane, at 49.5 GHz for the reference parameters.

I. INTRODUCTION

Altermagnets have two spin sublattices which are related by a crystal rotation rather than by translation or inversion [1–5]. Their momentum-dependent spin splitting is affected by the magnetic crystal symmetry, and d -wave altermagnetism has been observed in a metallic room-temperature material [6]. In the d -wave model used here, the two sublattices carry interchanged in-plane exchange stiffnesses, and the skyrmion consists of two elliptical sublattice textures whose long axes are perpendicular [7–10]. Skyrmions in antiferromagnets, ferrimagnets, ferromagnets, and magnetic multilayers could be manipulated by optical excitation, spin orbit torque, electric fields, and light induced strain [11–16]. For the altermagnet skyrmions, previous works have examined their its magnetic multipole, anisotropic Hall response, spin-transfer torques, current-driven dynamics, and interaction with pinning [7–9, 17–19].

At zero field, the two sublattice skyrmions carry opposite topological charges, and their net magnetic moment vanishes. The two perpendicular elliptical contours intersect along four directions. Their local mismatch elsewhere produces an alternating four-lobed pattern. This residual magnetization generates a magnetostatic field that acts on both sublattice textures.

For each sublattice component $\eta = A, B$, we define an effective skyrmion radius $R_\eta = \sqrt{a_\eta b_\eta}$, where a_η and b_η are the distances from the skyrmion center to its boundary along y and x , respectively. Our simulations show that an out-of-plane magnetic field enlarges

one sublattice skyrmion and shrinks the other. This opposite change of the two radii motivates the antisymmetric breathing coordinate studied here.

Breathing modes of single and antiferromagnetic skyrmions, as well as in-phase and antiphase modes of exchange-coupled skyrmions, are already known [20–22]. In a d -wave altermagnet, the exchange-anisotropy axes are interchanged between the two sublattices, making the two elliptical skyrmions perpendicular. We study how this altermagnetic geometry changes the equilibrium size of the skyrmion. To isolate the role of this orientation, we compare it with a reference in which the two ellipses have parallel long axes. The coordinate studied throughout the work is $\delta R = R_A - R_B$. In this work, the skyrmion cores on sublattices A and B point along $+z$ and $-z$, respectively, so a $+z$ field expands the A skyrmion and contracts the B skyrmion. At zero field, where $R_A = R_B$, the two perpendicular walls still do not coincide. The perpendicular geometry has a smaller radius than a parallel-ellipse reference built from the same exchange constants. We also generate a radius difference at zero field by applying a directional exchange modulation, which we use to represent uniaxial strain. We apply the same directional perturbation to the exchange couplings on both sublattices: all x -directed exchange coefficients are multiplied by $1 + \delta$, while all y -directed coefficients are multiplied by $1 - \delta$. Because the strong and weak exchange directions are interchanged between the two sublattices, the same perturbation makes one skyrmion more elongated and the other less elongated. As a result, a nonzero radius difference occurs. Beyond these static changes, the radius difference coordinate also supports a dynamical mode. After a short out-of-plane field pulse, the radius difference oscillates together with

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the uniform helicity, defined as the common rotation of the in-plane magnetization in the two domain walls. The helicity reaches an extremum when the radius difference crosses zero.

II. MODEL AND SIMULATED SKYRMION

The lattice Hamiltonian for a d -wave altermagnet is [7, 8],

$$\begin{aligned} \mathcal{H} = \sum_{i,j} \bigg[& -J_1 \mathbf{m}_{i,j}^A \cdot \mathbf{m}_{i+1,j}^A - J_2 \mathbf{m}_{i,j}^B \cdot \mathbf{m}_{i+1,j}^B \\ & - J_2 \mathbf{m}_{i,j}^A \cdot \mathbf{m}_{i,j+1}^A - J_1 \mathbf{m}_{i,j}^B \cdot \mathbf{m}_{i,j+1}^B \\ & - J_3 \mathbf{m}_{i,j}^A \cdot \mathbf{m}_{i,j}^B - g\mu_B \mathbf{B} \cdot (\mathbf{m}_{i,j}^A + \mathbf{m}_{i,j}^B) \\ & - D_0 (\mathbf{m}_{i,j}^A \times \mathbf{m}_{i+1,j}^A + \mathbf{m}_{i,j}^B \times \mathbf{m}_{i+1,j}^B) \cdot \hat{e}_y \\ & + D_0 (\mathbf{m}_{i,j}^A \times \mathbf{m}_{i,j+1}^A + \mathbf{m}_{i,j}^B \times \mathbf{m}_{i,j+1}^B) \cdot \hat{e}_x \\ & + K_0 [1 - (\mathbf{m}_{i,j}^A \cdot \hat{e}_z)^2] + K_0 [1 - (\mathbf{m}_{i,j}^B \cdot \hat{e}_z)^2] \bigg] \\ & + \mathcal{H}_{\text{ms}}. \end{aligned} \quad (1)$$

Here, $\mathbf{m}_{i,j}^\eta$ is the unit magnetization vector on sublattice $\eta = A, B$ at lattice site (i, j) . The parameters J_1 and J_2 are the nearest-neighbor exchange couplings within each sublattice. Their directions are interchanged between the two sublattices: sublattice A has exchange couplings J_1 and J_2 along x and y , respectively, whereas sublattice B has J_2 and J_1 . The parameter $J_3 < 0$ describes the antiferromagnetic exchange coupling between the two sublattices. The Zeeman term contains the Landé factor g , the Bohr magneton μ_B , and the applied magnetic field $\mathbf{B}$. D_0 is the strength of the interfacial Dzyaloshinskii–Moriya interaction (DMI) within each sublattice, and $K_0 > 0$ is the perpendicular easy-axis anisotropy along z . The term $\mathcal{H}_{\text{ms}}$ is the magnetostatic energy generated by the spatial distribution of the combined sublattice magnetization. The corresponding continuum energy density is

$$\begin{aligned} w = & A_1 |\partial_x \mathbf{m}_A|^2 + A_2 |\partial_y \mathbf{m}_A|^2 + A_2 |\partial_x \mathbf{m}_B|^2 + A_1 |\partial_y \mathbf{m}_B|^2 \\ & + K(1 - m_{Az}^2) + K(1 - m_{Bz}^2) \\ & + D(m_{Az} \partial_x m_{Ax} + m_{Az} \partial_y m_{Ay} - m_{Ax} \partial_x m_{Az} \\ & - m_{Ay} \partial_y m_{Az}) + D(m_{Bz} \partial_x m_{Bx} + m_{Bz} \partial_y m_{By} \\ & - m_{Bx} \partial_x m_{Bz} - m_{By} \partial_y m_{Bz}) \\ & - M_s \mathbf{B} \cdot \mathbf{m}_A - M_s \mathbf{B} \cdot \mathbf{m}_B + A_0 \mathbf{m}_A \cdot \mathbf{m}_B \\ & + w_{\text{ms}}. \end{aligned} \quad (2)$$

The continuum coefficients are related to the lattice parameters by $A_{1,2} = \frac{J_{1,2} a^2}{2v_c}$, $A_0 = -\frac{J_3}{v_c}$, $D = -\frac{D_0 a}{v_c}$, $K = \frac{K_0}{v_c}$, $M_s = \frac{g\mu_B}{v_c}$ where $v_c = a^2 t_c$ is the unit-cell volume, a the in-plane lattice constant, t_c the out-of-plane lattice period, and M_s the saturation magnetization of each sublattice. The magnetostatic term is

$$w_{\text{ms}}(\mathbf{r}) = -\frac{1}{2} \mathbf{M}(\mathbf{r}) \cdot \mathbf{H}_{\text{ms}}(\mathbf{r}), \quad \mathbf{M} = M_s(\mathbf{m}_A + \mathbf{m}_B),$$

where $\mathbf{H}_{\text{ms}}$ is the demagnetizing field expressed in tesla, with μ_0 included in its definition. For a film of thickness t , the magnetostatic energy is $E_{\text{ms}} = t \int d^2 r w_{\text{ms}}$. The field at each point is generated by the total magnetization throughout the film and acts on both sublattices. The thickness-averaged magnetostatic kernel is given in Supplemental Material, Sec. S1. The local anisotropy is $K = K_0/v_c$. The magnetization dynamics of each sublattice are described by the Landau–Lifshitz–Gilbert (LLG) equation

$$\frac{d\mathbf{m}_\eta}{dt} = -\frac{\gamma}{1+\alpha^2} [\mathbf{m}_\eta \times \mathbf{H}_{\eta,\text{eff}} + \alpha \mathbf{m}_\eta \times (\mathbf{m}_\eta \times \mathbf{H}_{\eta,\text{eff}})], \quad (3)$$

where $\gamma = 1.76 \times 10^{11} \text{ rad s}^{-1} \text{ T}^{-1}$ and α is the Gilbert damping parameter. The effective fields, expressed in tesla, are $\mathbf{H}_{\eta,\text{eff}}(\mathbf{r}) = -(M_s t)^{-1} \delta E / \delta \mathbf{m}_\eta(\mathbf{r})$, with $E = t \int d^2 r w$. We define

$$\bar{A} = \sqrt{A_1 A_2}, \quad \Delta A = A_1 - A_2. \quad (4)$$

Here the geometric mean $\bar{A}$ sets the overall exchange-stiffness scale, whereas ΔA measures the difference between the two in-plane directions. When $\Delta A = 0$, the exchange is isotropic and the exchange contrast of this two-sublattice model disappears. We minimize the total energy E in a $96 \times 96 \text{ nm}^2$ periodic cell on a 192×192 grid with spacing $h = 0.5 \text{ nm}$. The minimization procedure and the extraction of radii and ellipticities are described in Supplemental Material, Sec. S2. Following the parameterization in Ref. [8], we take $M_s = 580 \text{ kA/m}$, $A_1 = 2 \times 10^{-11} \text{ J/m}$, $A_2 = 1 \times 10^{-11} \text{ J/m}$, $A_0 = 2 \times 10^5 \text{ J/m}^3$, $D = 2.5 \times 10^{-3} \text{ J/m}^2$, $K = 6 \times 10^5 \text{ J/m}^3$, and film thickness $t = 1 \text{ nm}$. These parameters give $\bar{A} = \sqrt{2} \times 10^{-11} \text{ J/m} \simeq 1.41 \times 10^{-11} \text{ J/m}$ and the reference wall-width scale $\lambda = \sqrt{\bar{A}/K} \simeq 4.85 \text{ nm}$. We use $\rho = A_1/A_2$ for the exchange contrast and vary it at fixed $\bar{A}$.

Figures 1(a) and 1(b) show the out-of-plane magnetization of the two sublattices at zero field and $\rho = 2$. The two $m_z = 0$ contours are elliptical, with perpendicular long axes. The two radii are equal, $R_A = R_B = 3.87 \text{ nm}$, and the aspect ratio is $b_A/a_A = 1.40$. Panel (c) shows the sum $m_z^A + m_z^B$. It forms four lobes of alternating sign. The boundary of each sublattice component is its $m_z = 0$ contour. Where the A boundary lies outside the B boundary the local magnetization is positive, and where the reverse holds it is negative, which is the alternating pattern of panel (c). A compensated altermagnetic skyrmion therefore carries a spatially structured magnetization at zero field even though its net moment vanishes. In polar coordinates centered on the skyrmion, $r_\eta(\varphi)$ is the distance from the center to that contour in the direction φ , measured from the x axis. Panel (d) shows the angular variation of the mismatch between the two boundaries. The solid line gives $u(\varphi) = r_A(\varphi) - r_B(\varphi)$, obtained from the simulated $m_z = 0$ contours. The dashed line shows $(1.29 \text{ nm}) \cos 2\varphi$, with the amplitude extracted from these contours. The numerical $u(\varphi)$ is well approxi-

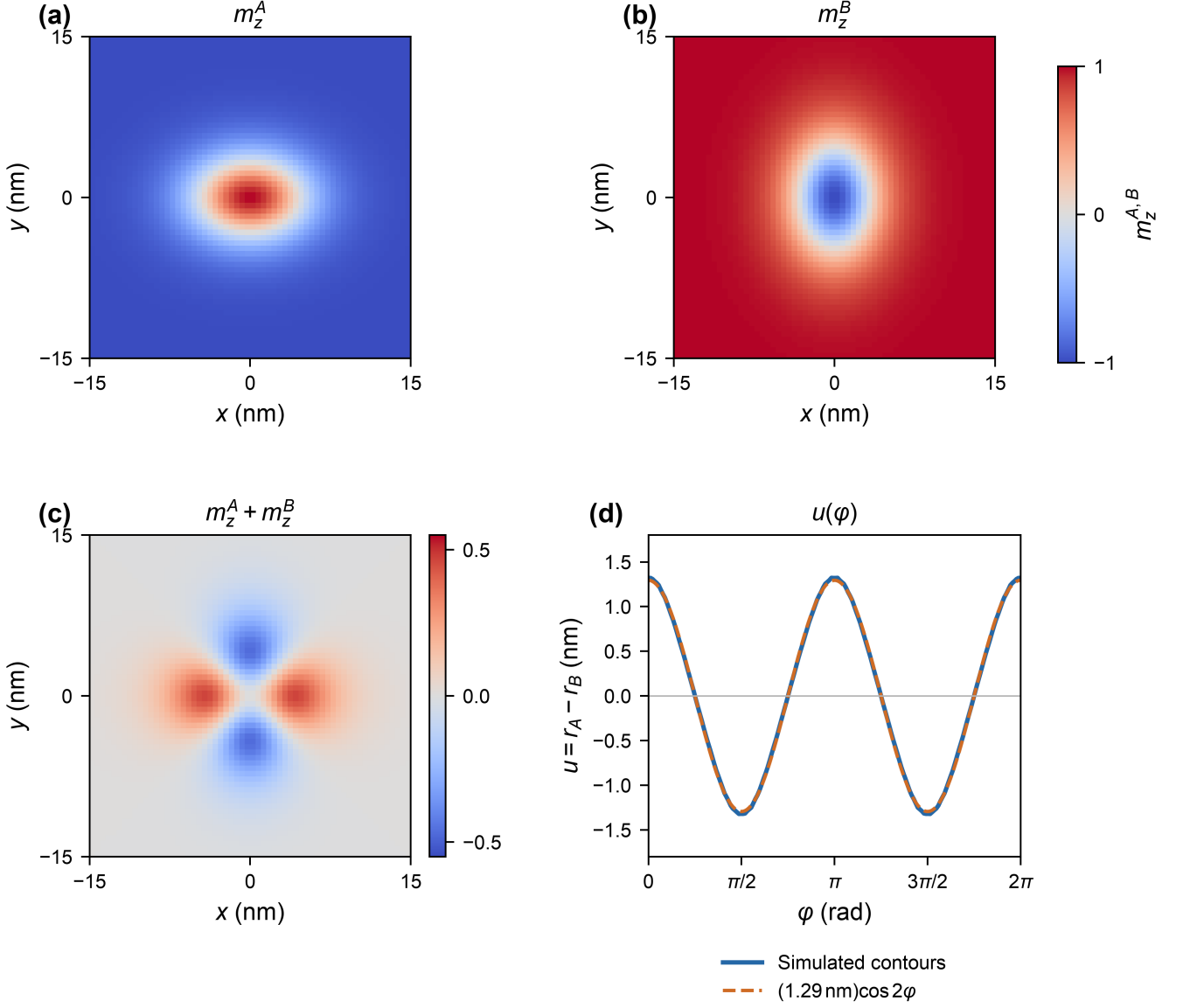


FIG. 1. Simulated d -wave altermagnetic skyrmion at zero field and $\rho = 2$. (a,b) Out-of-plane magnetization on sublattices A and B . (c) $m_z^A + m_z^B$. (d) $u(\varphi) = r_A(\varphi) - r_B(\varphi)$, with φ measured from the x axis. The solid curve uses the simulated contours. The dashed curve is $(1.29 \text{ nm}) \cos 2\varphi$.

mated by $(1.29 \text{ nm}) \cos 2\varphi$. The $\cos 2\varphi$ dependence shows that the leading contour mismatch changes sign every 90° , as expected for the two perpendicular elliptical contours. This angular mismatch produces the four-lobed local magnetization in Fig. 1(c).

At $B_z = +0.3 \text{ T}$, Fig. 2 shows that the A skyrmion expands from 3.87 to 4.58 nm and the B skyrmion contracts to 3.61 nm . The radius difference is $\delta R = 0.97 \text{ nm}$. Their mean radius increases to 4.09 nm , 5.67% above its zero-field value.

III. MAGNETIC FIELD AND RADIUS DIFFERENCE

A. Coordinates and changes with the field

At $B_z = 0$, the net moment $\mu_z = t \int d^2r M_z$ vanishes even though M_z is not zero everywhere. A small δR produces a contribution to μ_z proportional to δR , so the Zeeman energy contains a term proportional to $-B_z \delta R$. We describe the changes in a small field using the two radii and the two sublattice ellipticities.

We denote the boundary distances along x and y by

$$b_\eta = r_\eta(0), \quad a_\eta = r_\eta(\pi/2),$$

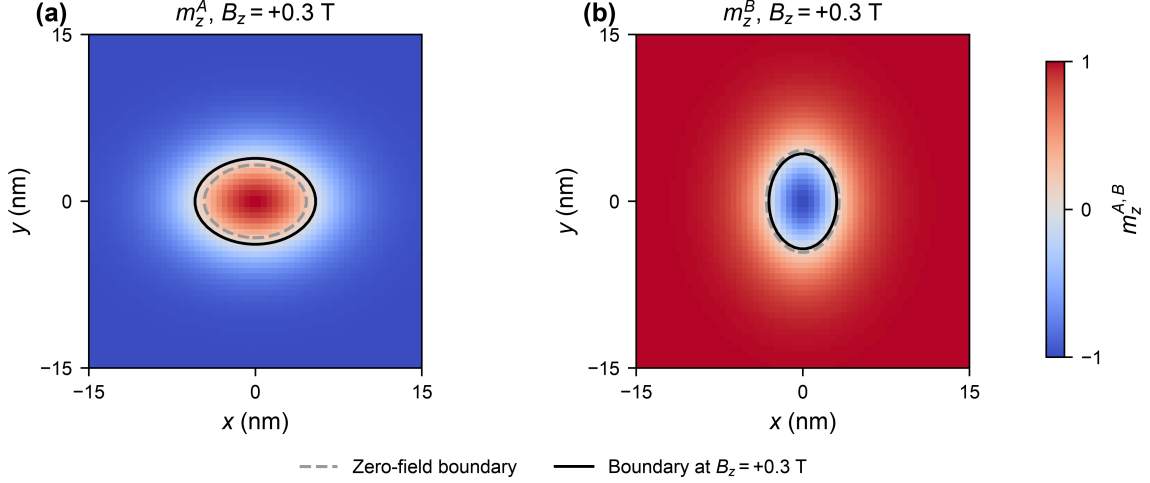


FIG. 2. Sublattice skyrmions at $B_z = +0.3$ T. The gray dashed curves show the $m_z = 0$ contours at $B_z = 0$, and the black curves show those at $B_z = +0.3$ T. (a) Sublattice A expands to $R_A = 4.58$ nm. (b) Sublattice B contracts to $R_B = 3.61$ nm.

and define

$$\varepsilon_\eta = \frac{b_\eta - a_\eta}{b_\eta + a_\eta}.$$

Positive ε_η denotes elongation along x , and negative ε_η denotes elongation along y . We use the following combinations of the two radii and ellipticities:

$$\begin{aligned} \bar{R} &= \frac{1}{2}(R_A + R_B), & \delta R &= R_A - R_B, \\ e_s &= \frac{1}{2}(\varepsilon_A - \varepsilon_B), & e_u &= \frac{1}{2}(\varepsilon_A + \varepsilon_B). \end{aligned} \quad (5)$$

The definitions give

$$R_A = \bar{R} + \delta R/2, \quad R_B = \bar{R} - \delta R/2,$$

so changing $\bar{R}$ at fixed δR changes both radii equally, whereas changing δR at fixed $\bar{R}$ changes them in opposite directions. Similarly,

$$\varepsilon_A = e_s + e_u, \quad \varepsilon_B = -e_s + e_u.$$

At zero field, $e_u = 0$ and $\varepsilon_A = -\varepsilon_B$. At fixed e_s , a change Δe_u gives $\Delta \varepsilon_A = \Delta \varepsilon_B = \Delta e_u$.

Reflection about $x = y$, combined with sublattice exchange, acts as

$$\begin{aligned} \mathcal{P}: \quad \mathbf{m}_A(x, y) &\rightarrow -\mathcal{R} \mathbf{m}_B(y, x), \\ \mathbf{m}_B(x, y) &\rightarrow -\mathcal{R} \mathbf{m}_A(y, x), \end{aligned} \quad (6)$$

Here $\mathcal{R}(m_x, m_y, m_z) = (m_y, m_x, m_z)$. At zero field, all energy terms, including magnetostatics, are unchanged under $\mathcal{P}$. At fixed nonzero B_z , $\mathcal{P}$ reverses the Zeeman term. If B_z is reversed simultaneously with $\mathcal{P}$, the Zeeman term is restored and the full energy is unchanged. The term-by-term verification is given in Supplemental Material, Sec. S3.

The reflection sends φ to $\pi/2 - \varphi$. Together with sublattice exchange, it gives $R_A \leftrightarrow R_B$ and $\varepsilon_A \leftrightarrow -\varepsilon_B$. Thus,

$$\begin{aligned} (\bar{R}, e_s) &\rightarrow (\bar{R}, e_s), & \mathcal{P}\text{-even}, \\ (\delta R, e_u, B_z) &\rightarrow (-\delta R, -e_u, -B_z), & \mathcal{P}\text{-odd}. \end{aligned} \quad (7)$$

For fixed values of $(\bar{R}, e_s, \delta R, e_u)$ and B_z , we define $E(\bar{R}, e_s, \delta R, e_u; B_z)$ as the minimum of the full spin energy over configurations with these collective coordinates. Eq. (7) makes this energy invariant under simultaneous reversal of $(\delta R, e_u, B_z)$. At quadratic order, $\mathcal{P}$ permits $B_z \delta R$, $B_z e_u$, and $\delta R e_u$, but excludes terms linear in δR , e_u , or B_z individually. We therefore write

$$\begin{aligned} E &= E_0(\bar{R}, e_s; B_z^2) - B_z(\Lambda_R \delta R + \Lambda_{e_u} e_u) \\ &\quad + \frac{1}{2} \kappa_0 \delta R^2 + \frac{1}{2} \kappa_{e_u} e_u^2 + \zeta \delta R e_u + \mathcal{O}(4). \end{aligned} \quad (8)$$

At fixed $\bar{R}$ and e_s , the coefficients are evaluated at $B_z = \delta R = e_u = 0$. The coefficient κ_0 is the second derivative of E with respect to δR while $e_u = 0$, and κ_{e_u} is the second derivative with respect to e_u while $\delta R = 0$. They determine the quadratic energy costs of the two deformations. The term $\zeta \delta R e_u$ couples the radius difference to the uniform ellipticity. The coefficients Λ_R and Λ_{e_u} give the terms proportional to δR and e_u in the net moment at zero field. The term E_0 is independent of these two coordinates and is even in B_z . Here $\mathcal{O}(4)$ denotes fourth and higher total order in $(\delta R, e_u, B_z)$, with $\bar{R}$ and e_s held fixed.

A small B_z changes δR and e_u linearly, whereas the changes in $\bar{R}$ and e_s begin at order B_z^2 :

$$\bar{R}(B_z) = \bar{R}(0) + \mathcal{O}(B_z^2), \quad e_s(B_z) = e_s(0) + \mathcal{O}(B_z^2).$$

Minimizing the quadratic energy in Eq. (8) with respect

to e_u and δR gives, to first order in B_z ,

$$\begin{aligned} e_u &= \frac{\Lambda_{e_u} B_z - \zeta \delta R}{\kappa_{e_u}}, \\ \delta R &= \frac{\Lambda_R - \zeta \Lambda_{e_u} / \kappa_{e_u}}{\kappa} B_z \equiv \chi B_z, \\ \kappa &= \kappa_0 - \frac{\zeta^2}{\kappa_{e_u}}. \end{aligned} \quad (9)$$

The first equation contains a term $\Lambda_{e_u} B_z / \kappa_{e_u}$ from the field and a term $-\zeta \delta R / \kappa_{e_u}$ from the coupling to δR . To determine the zero-field energy cost of a small δR , we fix $\bar{R}$, e_s , and δR and minimize over e_u . For $\kappa_{e_u} > 0$, the quadratic energy gives

$$e_u = -\frac{\zeta}{\kappa_{e_u}} \delta R.$$

After this minimization, the energy increase relative to the $\delta R = 0$ state is

$$\Delta E(\delta R) = \frac{1}{2} \kappa \delta R^2 + O(\delta R^4), \quad \kappa = \kappa_0 - \frac{\zeta^2}{\kappa_{e_u}}.$$

For $\kappa_{e_u} > 0$, allowing e_u to change lowers the energy by $\zeta^2 \delta R^2 / (2\kappa_{e_u})$ relative to holding $e_u = 0$. Thus $\kappa \leq \kappa_0$, with equality when $\zeta = 0$. The quadratic energy in $(\delta R, e_u)$ is positive when $\kappa_{e_u} > 0$ and $\kappa > 0$. At $\kappa = 0$, higher-order terms are needed to determine stability. The slope $\chi = (d\delta R / dB_z)_0$ measures the radius difference produced by a small field. Using the states at $B_z = \pm 0.05$ T, we obtain

$$\chi \simeq \frac{\delta R(0.05 \text{ T}) - \delta R(-0.05 \text{ T})}{0.10 \text{ T}} = 3.08 \text{ nm/T}. \quad (10)$$

The symmetric differences are described in Supplemental Material, Sec. S4. Figure 3(a) shows the increase of δR with B_z . In Fig. 3(b), reversing B_z leaves e_s unchanged and reverses e_u , consistent with Eq. (7). Over the plotted interval, $|e_u| < 10^{-3}$, compared with $e_s(0) \simeq 1.68 \times 10^{-1}$. A small positive B_z therefore increases R_A and decreases R_B while changing the signed ellipticities only slightly. For the symmetric field pairs $B_z = \pm B_1$ with $B_1 = 0.025$ and 0.05 T, the differences $[e_u(B_1) - e_u(-B_1)] / (2B_1)$ are negative. At larger positive fields, e_u changes sign. The additional calculations are reported in Supplemental Material, Sec. S4.

In summary, δR is linear in B_z , while the change in $\bar{R}$ begins at order B_z^2 . The uniform ellipticity e_u can also change linearly with B_z , but it remains below 10^{-3} over the plotted range, and e_s has no linear change. Thus, the first-order size change produced by the field is the radius difference δR , rather than a common change of the two radii. This is why we use δR as the radial coordinate for the field-driven oscillation studied in Sec. VI.

B. From the contours to the magnetic moment

We now relate the contour difference $u(\varphi)$ to the local magnetization and the net moment, explaining why

a uniform B_z couples to δR . We approximate m_{Az} and $-m_{Bz}$ by the same decreasing radial profile f , with $f(0) = +1$, $f(1) = 0$, and $f(\infty) = -1$:

$$m_{Az} = f\left(\frac{r}{r_A(\varphi)}\right), \quad m_{Bz} = -f\left(\frac{r}{r_B(\varphi)}\right). \quad (11)$$

To first order in ellipticity, the contour is

$$r_\eta(\varphi) \simeq R_\eta[1 + \varepsilon_\eta \cos 2\varphi]. \quad (12)$$

The expansion through second order is derived in Supplemental Material, Sec. S5. Subtracting the two contours gives

$$\begin{aligned} u(\varphi) &= r_A(\varphi) - r_B(\varphi) \\ &\simeq \delta R + (2\bar{R}e_s + \delta R e_u) \cos 2\varphi. \end{aligned} \quad (13)$$

Expanding $f(r/r_A)$ and $f(r/r_B)$ to first order about $r_A = r_B = \bar{R}$ gives

$$\begin{aligned} M_z(r, \varphi) &= M_s(m_{Az} + m_{Bz}) \simeq M_s p(r) u(\varphi), \\ p(r) &= -\frac{r}{\bar{R}^2} f'\left(\frac{r}{\bar{R}}\right) > 0. \end{aligned} \quad (14)$$

The radial factor $p(r)$ is concentrated near the wall, while $u(\varphi)$ determines the angular sign of M_z . The derivation is given in Supplemental Material, Sec. S5.

At $B_z = 0$, $u(\varphi) \simeq 2\bar{R}e_s \cos 2\varphi$, giving four lobes of M_z with alternating signs and zero net moment [7, 8]. At $\rho = 2$, $u(\varphi) \simeq (1.29 \text{ nm}) \cos 2\varphi$, with an amplitude of about 0.27λ for $\lambda = 4.85 \text{ nm}$. Angular integration gives

$$\mu_z = t \int d^2r M_z = 2\pi M_s t \delta R \int_0^\infty r p(r) dr, \quad (15)$$

to first order, with $\Lambda_R = 2\pi M_s t \int_0^\infty r p(r) dr$. For a circular contour with $f(\xi) = +1$ for $\xi < 1$ and $f(\xi) = -1$ for $\xi > 1$,

$$\Lambda_R = 4\pi M_s t \bar{R}. \quad (16)$$

For elliptical contours and the scaled profile in Eq. (11), each core integral is proportional to R_η^2 , independently of ε_η . Consequently, $\mu_z \propto R_A^2 - R_B^2 = 2\bar{R}\delta R$. The higher magnetic moments of the out-of-plane and in-plane magnetization are derived in Supplemental Material, Sec. S6.

A uniform B_z couples to the integral of M_z , giving

$$E_Z = -B_z t \int d^2r M_z = -B_z \mu_z = -\Lambda_R B_z \delta R. \quad (17)$$

The $\cos 2\varphi$ term integrates to zero. Changing e_u at fixed R_A and R_B leaves μ_z unchanged, giving $\Lambda_{e_u} = 0$. The calculations in Supplemental Material, Sec. S7 give a nonzero estimate of Λ_{e_u} for the numerical profiles. Since $\Lambda_R > 0$ for the chosen core polarities, a positive B_z favors a positive δR .

Hence, the altermagnetic skyrmion can have a nonzero local magnetization while its net moment vanishes. $\cos 2\varphi$ part of $u(\varphi)$ produces the four-lobed zero-field pattern, whereas the angle-independent part δR produces a net moment to first order.

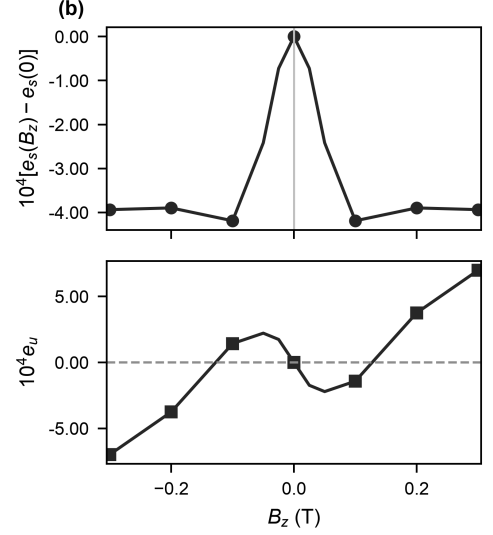
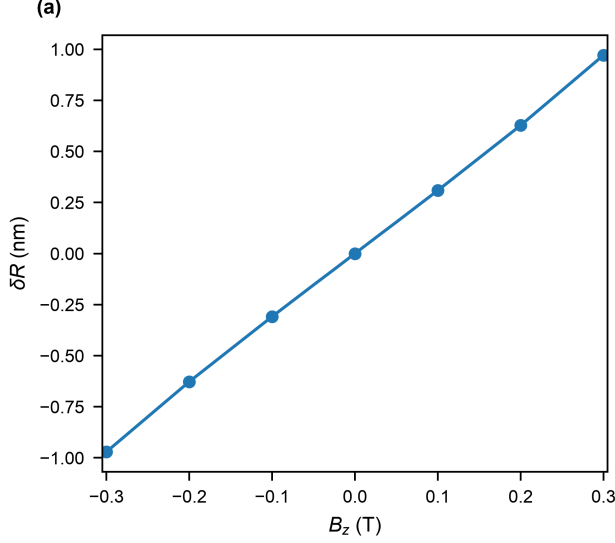


FIG. 3. Changes in the radii and ellipticities at $\rho = 2$. (a) δR versus B_z . (b) $10^4[e_s(B_z) - e_s(0)]$ and $10^4e_u(B_z)$, showing the even and odd dependences, respectively.

IV. RADIUS DIFFERENCE INDUCED BY EXCHANGE MODULATION

To study a radius difference generated without a magnetic field, we model strain by applying the same directional exchange modulation to both sublattices at $B_z = 0$:

$$\begin{aligned} (A_{Ax}, A_{Ay}) &= (A_1(1+\delta), A_2(1-\delta)), \\ (A_{Bx}, A_{By}) &= (A_2(1+\delta), A_1(1-\delta)), \end{aligned} \quad (18)$$

Here δ is the dimensionless relative change of the exchange coefficients: the coefficients along x change by $+\delta$ and those along y by $-\delta$, relative to their unperturbed values. Its relation to lattice strain requires the material dependence of the exchange coefficients. The anisotropy and DMI are held fixed. The factors are the same on both sublattices, but the absolute changes differ when $A_1 \neq A_2$. For both sublattices, the geometric mean is $\bar{A}\sqrt{1-\delta^2} = \bar{A}[1 - \delta^2/2 + O(\delta^4)]$. For a fixed dimensionless Néel profile and aspect ratio b_η/a_η , minimization of the local single-sublattice energy with respect to R_η gives

$$R_\eta = R_0 \left(\sqrt{\frac{b_\eta}{a_\eta}} + \sqrt{\frac{a_\eta}{b_\eta}} \right), \quad (19)$$

where $R_0 \propto D/K$ depends on the assumed profile. This estimate omits intersublattice exchange and magneto-statics. Its derivation is given in Supplemental Material, Sec. S8.

Minimizing only the exchange energy gives $b_A/a_A = \sqrt{\rho}$ and $b_B/a_B = 1/\sqrt{\rho}$. The modulation multiplies both ratios by $[(1+\delta)/(1-\delta)]^{1/2}$. Equation (19) gives an increase of R_A and a decrease of R_B . To first order,

$$\delta R \simeq R_0(\rho^{1/4} - \rho^{-1/4})\delta, \quad (20)$$

for the assumed local profile. We define $\chi_\epsilon = (d\delta R/d\delta)_0$ as the slope of the radius difference with respect to the exchange modulation. We obtain it by minimizing the full energy at each δ and fitting $\delta R(\delta)$ over $|\delta| \leq 0.02$. The derivation and numerical values are given in Supplemental Material, Sec. S8.

Figure 4(a) shows $\delta R(\delta)$ for $-0.06 \leq \delta \leq 0.06$. Figure 4(b) shows χ_ϵ as a function of ρ . It is 3.85×10^{-1} nm at $\rho = 2$ and 1.19 nm at $\rho = 4$. At $\rho = 1$, the two sublattices have identical exchange tensors and $\chi_\epsilon = 0$, consistent with the numerical result. As $|\delta|$ increases, the curves in Fig. 4(a) depart from the linear relation $\delta R = \chi_\epsilon \delta$.

The interchanged exchange directions therefore allow the same directional modulation on both sublattices to change their radii oppositely at zero field. For the modulation studied here, χ_ϵ vanishes when the sublattice exchange tensors are identical and increases over the sampled $\rho > 1$. This gives a way to change the internal radius difference through the exchange coefficients, distinct from the magnetic-field coupling, which remains nonzero at $\rho = 1$.

V. EQUILIBRIUM SIZE AND RADIUS DIFFERENCE

We test how the relative orientation of the two sublattice textures affects the size and field-induced radius difference of an altermagnetic skyrmion. We denote the arrangement with (A_1, A_2) along (x, y) on sublattice A and (A_2, A_1) on sublattice B by AM. Its two sublattice skyrmions have perpendicular long axes. In the reference PAR, both sublattices have (A_1, A_2) along (x, y) , and the long axes are parallel. We compare their zero-field

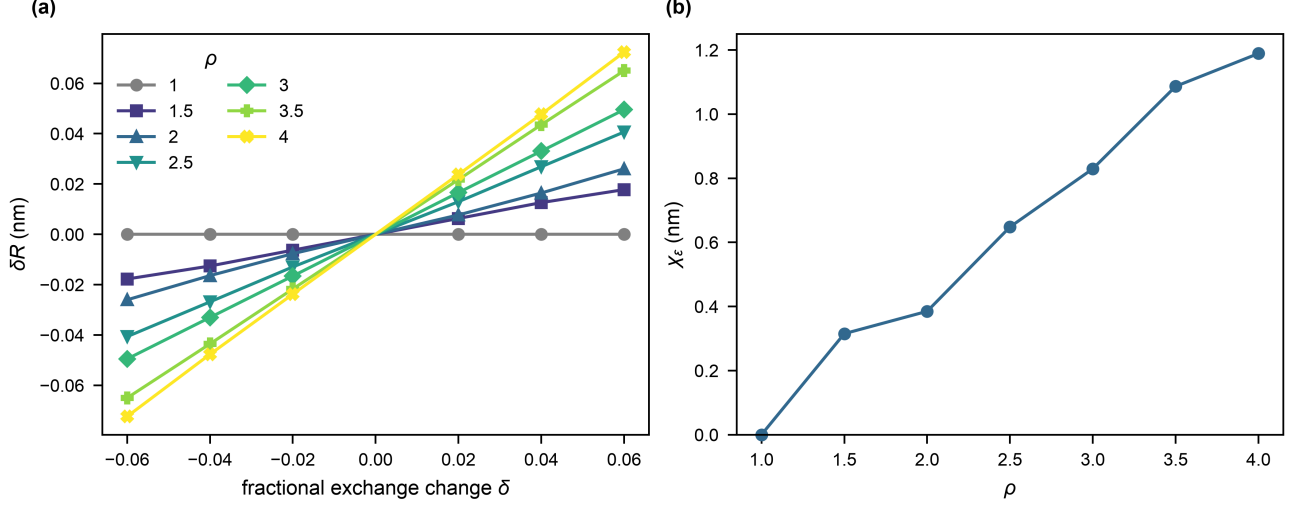


FIG. 4. Radius difference produced by the exchange modulation at $B_z = 0$. All coefficients along x are multiplied by $1 + \delta$ and those along y by $1 - \delta$ on both sublattices. The unperturbed $\bar{A} = \sqrt{A_1 A_2}$ and the other parameters are fixed as in Sec. II. (a) δR versus δ for the listed $\rho = A_1/A_2$. Reversing δ reverses δR , which vanishes at $\rho = 1$. (b) $\chi_\epsilon = (d\delta R/d\delta)_0$ versus ρ .

mean radii $\bar{R}$ and small-field slopes $\chi = (d\delta R/dB_z)_0$. For AM at $\rho = 2$, we then calculate the zero-field energy of unequal-radius states to construct the radial part of the oscillation model.

A. Perpendicular and parallel exchange axes

All coefficients other than the assignment of A_1 and A_2 to the two sublattices are unchanged, as are the film thickness and magnetostatic calculation. PAR is a collinear antiferromagnetic reference with anisotropic in-plane exchange. The subscript $\parallel$ denotes PAR. At $\rho = 1$, the two models coincide.

Figure 5(a) and Table I compare the zero-field radii. AM has the smaller $\bar{R}$ at every sampled $\rho > 1$. At $\rho = 4$, $\bar{R}_{\text{AM}} = 3.53$ nm and $\bar{R}_{\parallel} = 5.73$ nm.

Figure 5(b) compares estimates of χ obtained from the states at $B_z = \pm 0.05$ T on the $h = 0.5$ nm grid. At $\rho = 4$, χ is 5.53% smaller in AM than in PAR. The moment change per unit radius difference, $\Lambda_{\text{path}} = \Delta\mu_z/\Delta\delta R$, is 28.36% smaller. Here $\Delta X = X(0.05 \text{ T}) - X(-0.05 \text{ T})$, and both percentages use PAR as the reference. Equation (9) shows that χ depends on both the magnetic coupling and κ . The states and differences are specified in Supplemental Material, Sec. S4.

At $\rho = 2$, reducing h from 0.50 to 0.25 nm in the same 96 nm cell changes $\chi_{\parallel} - \chi_{\text{AM}}$ from 1.83×10^{-2} to 6.17×10^{-2} nm/T, using $B_z = \pm 0.05$ T. PAR has the larger value on both grids, while the magnitude of the difference remains sensitive to h .

The local wall model in Supplemental Material, Sec. S9 identifies an energy cost of the noncoincident contours. Here $E_{\text{inter}} = A_0 t \int d^2 r [1 + \mathbf{m}_A \cdot \mathbf{m}_B]$ is the excess intersublattice energy relative to exactly antiparallel spins.

For two walls of the same width λ , the model gives

$$E_{\text{inter}} \simeq A_0 t \lambda \oint \bar{R} d\varphi F\left(\frac{u(\varphi)}{\lambda}\right),$$

where F is even, vanishes at zero, and increases with the magnitude of its argument for the profiles used there. In AM, $u(\varphi) \simeq 2\bar{R}e_s \cos 2\varphi$ already at $\delta R = 0$. Increasing $\bar{R}$ while keeping the shape and λ fixed raises this intersublattice energy. In the symmetric PAR state, $u = 0$ and the two magnetizations cancel at every point, so this energy is zero. The equilibrium radii in Fig. 5(a) are obtained by minimizing all terms in Eq. (2).

AM also has a nonzero magnetostatic energy. At $\rho = 2$, the contributions from magnetization in the film plane and along z are 8.96×10^{-22} J and 2.18×10^{-21} J, respectively. Both are included in the numerical minimization.

Across $\rho = 1$ to 4, $\bar{R}_{\text{AM}}$ decreases by about 14.09%, whereas $\bar{R}_{\parallel}$ increases by about 39.37%. Over the same scan, the small-field slopes increase in both geometries. Thus the AM skyrmion becomes smaller as ρ increases, while a given small field produces a larger radius difference.

These results show that the size of an altermagnetic skyrmion is controlled not only by the properties of each sublattice texture, but also by their relative geometry.

B. Energy of unequal radii

To construct the radial part of the oscillation model, we calculate a family of states at $B_z = 0$ and $\rho = 2$ in AM. For each prescribed

$$q = \sqrt{S_A/\pi} - \sqrt{S_B/\pi}, \quad S_\eta = \frac{1}{2} \int d^2 r [1 + s_\eta m_{\eta z}],$$

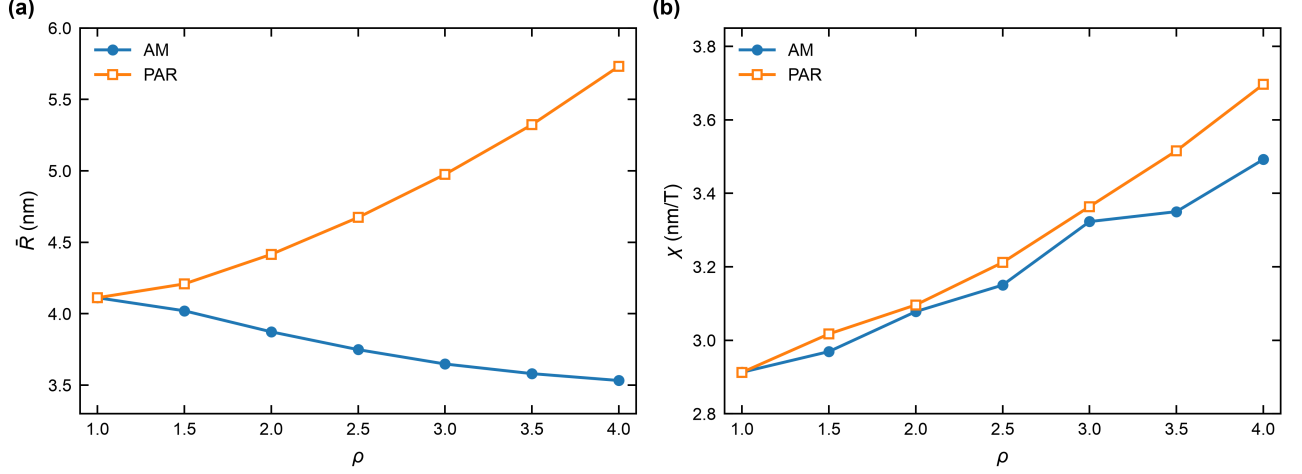


FIG. 5. AM and PAR compared at fixed $\bar{A}$ versus $\rho = A_1/A_2$. (a) Mean radius $\bar{R}$ at $B_z = 0$. (b) $\chi = (d\delta R/dB_z)_0$ using the symmetric difference $\chi \simeq \frac{\delta R(B_1) - \delta R(-B_1)}{2B_1}$, $B_1 = 0.05$ T. Both geometries use the same 96 nm cell, $h = 0.5$ nm grid, and remaining parameters. Blue filled circles denote AM and orange open squares denote PAR.

TABLE I. Zero-field mean radii and small-field slopes for AM and PAR at fixed $\bar{A} = \sqrt{2} \times 10^{-11}$ J/m. The radii use $R_\eta = \sqrt{a_\eta b_\eta}$. Each χ is estimated as $[\delta R(B_1) - \delta R(-B_1)]/(2B_1)$, with $B_1 = 0.05$ T. The entries use $h = 0.5$ nm in a 96 nm periodic cell. The additional grid comparison at $\rho = 2$ is reported in Sec. V A. AM and PAR coincide at $\rho = 1$.

ρ	$\bar{R}_{\text{AM}}$ (nm)	$\bar{R}_{\parallel}$ (nm)	χ_{AM} (nm/T)	$\chi_{\parallel}$ (nm/T)
1	4.11	4.11	2.91	2.91
1.5	4.02	4.21	2.97	3.02
2	3.87	4.42	3.08	3.10
2.5	3.75	4.67	3.15	3.21
3	3.65	4.98	3.32	3.36
3.5	3.58	5.32	3.35	3.52
4	3.53	5.73	3.49	3.70

with $s_A = +1$ and $s_B = -1$, we minimize the energy over the remaining spin variables. We then measure $\delta R = \sqrt{a_A b_A} - \sqrt{a_B b_B}$ from the resulting contours.

The states in Fig. 6 retain both cores for $|\delta R| \lesssim 2.805$ nm. Their energy is lowest at $\delta R = 0$. A fit of $E = E_c + \frac{1}{2}\kappa_{\text{con}}\delta R^2$ over $|\delta R| \leq 0.8$ nm gives $\kappa_{\text{con}} = 12.17$ mJ/m². The intercept E_c is fitted independently. In Sec. III A, κ is defined by minimizing the energy over e_u at fixed δR . Here κ_{con} is fitted to states minimized at fixed q . We use this family and κ_{con} for the radial part of the model in Sec. VI. The constraints and fit are described in Supplemental Material, Sec. S10.

These states identify an internal deformation of the altermagnetic skyrmion in which the two sublattice radii become unequal while both cores remain present. Along the selected fixed- q family, a small δR raises the zero-field energy by $\frac{1}{2}\kappa_{\text{con}}\delta R^2$. This energy cost supplies the radial restoring term in the oscillation model of Sec. VI.

VI. COUPLED OSCILLATIONS OF RADIUS DIFFERENCE AND HELICITY

We apply a short perpendicular field pulse and record $\delta R(t)$ after the field is removed. We also measure the helicity of each sublattice relative to the radial direction. The two equilibrium helicities differ by π . After subtracting each sublattice's zero-field value, we denote the helicity deviations by ϕ_A and ϕ_B . We define the uniform and staggered deviations as $\bar{\phi} = (\phi_A + \phi_B)/2$ and $\phi_- = (\phi_A - \phi_B)/2$. Their extraction from the spins and the time-integration and fitting procedures are described in Supplemental Material, Sec. S11.

We apply $B_z = 0.05$ T for 10 ps, set $B_z = 0$, and integrate the LLG equations for 2 ns with $\alpha = 10^{-3}$. At $\rho = 2$, the fitted frequency is 49.52 GHz and the ratio of the fitted oscillation amplitudes is $|\bar{\phi}|/|\delta R| = 1.57 \times 10^8$ rad/m. With the cosine convention used in the fit, $\bar{\phi}$ leads δR by 90.01° , approximately one quarter of a period. Figure 7(b) shows the phase difference between δR and $\bar{\phi}$. Figure 7(a) compares the fitted frequencies in AM and PAR as ρ is varied. With the other parameters fixed, the

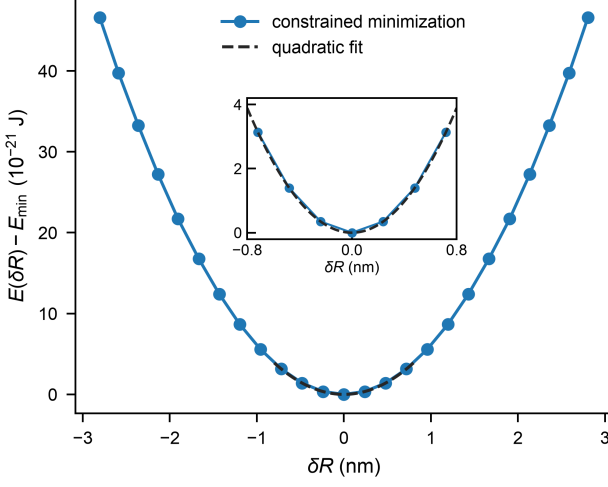


FIG. 6. Energy of AM states minimized at fixed q , plotted against the measured δR , at $B_z = 0$ and $\rho = 2$. Energies are measured relative to the zero-field state at $\delta R = 0$. All plotted states retain both cores. The dashed curve is a fit of $E = E_c + \frac{1}{2}\kappa_{\text{con}}\delta R^2$ over $|\delta R| \leq 0.8$ nm. The inset shows the interval used in the fit.

AM frequency decreases from 50.56 GHz at $\rho = 1$ to 46.51 GHz at $\rho = 4$, where the PAR value is 47.98 GHz.

Under $\mathcal{P}$, $\bar{\phi}$ changes sign and ϕ_- does not. Among the retained coordinates, only $(\delta R, e_u, \bar{\phi})$ can be excited to first order in the amplitude of a uniform B_z pulse. The oscillating parts of $\bar{R}$, e_s , and ϕ_- vanish at this order. At the δR peak, the Fourier amplitudes of $\bar{R}$ and ϕ_- are much smaller than those of δR and $\bar{\phi}$, respectively. These amplitudes, together with that of e_u , are reported in Supplemental Material, Sec. S11.

We approximate the motion by the family of states minimized at fixed q in Sec. V B, labelled by the measured δR , together with a common additional spin rotation $\bar{\phi}$ about z . This defines a variational model with coordinates δR and $\bar{\phi}$. We evaluate its Berry contribution as follows. Define $I_\eta = \int d^2r (1 + s_\eta \cos \theta_\eta)$, where θ_η is the polar angle of $\mathbf{m}_\eta$, $s_A = +1$, and $s_B = -1$. For spatially uniform increments ϕ_A and ϕ_B of the spin azimuth relative to the static profiles, the Berry contribution is

$$S_B = \frac{M_s t}{\gamma} \int dt \left[-I_A \dot{\phi}_A + I_B \dot{\phi}_B \right]. \quad (21)$$

The integrands in I_A and I_B vanish in the respective backgrounds. The sign convention is derived in Supplemental Material, Sec. S12. Substituting $\phi_{A,B} = \bar{\phi} \pm \phi_-$ gives

$$S_B = -\frac{M_s t}{\gamma} \int dt \left[(I_A - I_B) \dot{\bar{\phi}} + (I_A + I_B) \dot{\phi}_- \right]. \quad (22)$$

The definitions give $M_s t (I_A - I_B) = \mu_z$, so $I_A - I_B = 0$ in the symmetric state. Its first-order change along the fixed- q family gives the Berry term $-P \delta R \dot{\bar{\phi}}$. The

coefficient P , evaluated at $q = \delta R = 0$, is

$$P = \frac{M_s t}{\gamma} \left. \frac{d(I_A - I_B)}{d\delta R} \right|_0 \quad (23)$$

Equivalently, $P = \gamma^{-1} d\mu_z / d\delta R$, evaluated along the same states minimized at fixed q . The sum $I_A + I_B$ multiplies $\dot{\phi}_-$ and changes when both radii change together. The analogous pairing of radius and spin angle has been used for a single skyrmion and coupled layers [20, 22].

The frequency also requires the energy cost of changing $\bar{\phi}$. We set $\phi_A = \phi_B = \bar{\phi}$ while keeping the contours and polar angle profiles fixed, and define this cost as $\frac{1}{2}\kappa_{\bar{\phi}}\bar{\phi}^2$. This rotation leaves the local exchange, anisotropy, and intersublattice terms unchanged. It changes the DMI and magnetostatic energies. For two circular Néel walls with width negligible compared with $\bar{R}$, the estimate is

$$\kappa_{\bar{\phi}} \simeq 4\pi^2 D \bar{R} t + \kappa_{\bar{\phi}}^{\text{ms}}, \quad (24)$$

We obtain $\kappa_{\bar{\phi}}$ from the second-order energy change under a common rotation of the spins in the zero-field state. At $\rho = 2$, the fit gives $\kappa_{\bar{\phi}} = 4.08 \times 10^{-19}$ J. The separate DMI and magnetostatic contributions are given in Supplemental Material, Sec. S13.

The product $\delta R \dot{\bar{\phi}}$ is unchanged under $\mathcal{P}$. A mirror in x combined with time reversal leaves δR unchanged and reverses $\bar{\phi}$, which forbids this term. The verification is given in Supplemental Material, Sec. S14. For this trial family, the quadratic Lagrangian without damping is

$$L_{\text{var}} = -P \delta R \dot{\bar{\phi}} - \frac{1}{2} \kappa_{\text{con}} \delta R^2 - \frac{1}{2} \kappa_{\bar{\phi}} \bar{\phi}^2.$$

The equations are $P \delta \dot{R} = \kappa_{\bar{\phi}} \bar{\phi}$ and $P \dot{\bar{\phi}} = -\kappa_{\text{con}} \delta R$. They give $m^* \delta \ddot{R} = -\kappa_{\text{con}} \delta R$, with

$$m^* = \frac{P^2}{\kappa_{\bar{\phi}}}, \quad (25)$$

and $\omega_0 = \sqrt{\kappa_{\text{con}} \kappa_{\bar{\phi}}} / |P|$. Here m^* is the effective mass associated with the radius-difference coordinate δR . The corresponding frequency is $f_0 = \omega_0 / (2\pi)$. The circular-wall estimate of m^* for $\lambda / \bar{R} \ll 1$ is derived in Supplemental Material, Sec. S14. The role of internal spin deformations in skyrmion inertia is discussed in Refs. [23–25].

For AM at $\rho = 2$, the states minimized at fixed q give $P = 2.12 \times 10^{-22}$ J s/m and $\kappa_{\text{con}} = 12.17$ mJ/m². Together with $\kappa_{\bar{\phi}}$ from the common spin rotation, the unrounded coefficients give $f_0 = 53.00$ GHz and $|\dot{\bar{\phi}}|/|\delta R| = 1.73 \times 10^8$ rad/m, compared with 49.52 GHz and 1.57×10^8 rad/m from LLG. The frequency and amplitude ratio are 7.03% and 9.58% above the LLG values, respectively, using the unrounded coefficients. The same coefficients give $m^* = 1.10 \times 10^{-25}$ kg. The dependence of e_u on the chosen family and its small oscillation in LLG are examined in Supplemental Material, Sec. S15.

The altermagnetic skyrmion therefore supports a field-excitable internal mode in which its two sublattice radii

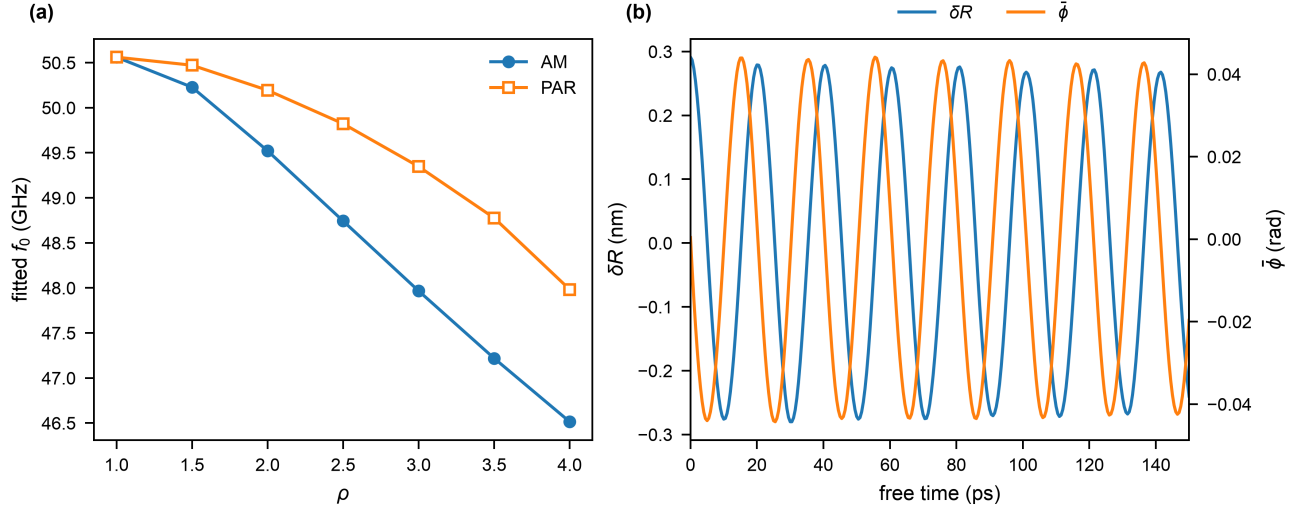


FIG. 7. Oscillations after a perpendicular field pulse. (a) Fitted frequency versus $\rho = A_1/A_2$ for AM and PAR. (b) $\delta R(t)$ on the left axis and $\bar{\phi}(t)$ in radians on the right axis for AM at $\rho = 2$. The pulse has amplitude 0.05 T and duration 10 ps. The subsequent calculation uses $B_z = 0$ and $\alpha = 10^{-3}$. Both $\delta R(t)$ and $\bar{\phi}(t)$ in (b) are measured from their zero-field equilibrium values.

oscillate oppositely and the uniform helicity is approximately one quarter of a period ahead of δR . For the parameters studied here, this mode lies near 50 GHz. Its coupling to a perpendicular field suggests studying the mode by excitation with an oscillating out-of-plane field at comparable frequencies.

VII. CONCLUSION

The altermagnetic skyrmion studied here has an internal size coordinate $\delta R = R_A - R_B$ that can be controlled by a uniform out-of-plane magnetic field. A small δR contributes linearly to the net magnetic moment and couples to B_z through the Zeeman term $-\Lambda_R B_z \delta R$. Near the zero-field equilibrium, δR and e_u can vary linearly with B_z , whereas the changes in $\bar{R}$ and e_s begin at order B_z^2 . The field therefore changes the relative sizes of the two sublattice textures without changing their mean radius to first order.

The equilibrium size also depends on the relative orientation of the two sublattice textures. In AM, equal radii do not make the perpendicular contours coincide. Their difference produces four lobes of local M_z whose integral vanishes at zero field. In the local wall model, this contour difference gives an excess intersublattice energy that increases with $\bar{R}$ at fixed ellipticity and wall width. The full-energy calculations give a smaller equilibrium radius in AM than in PAR at the sampled $\rho > 1$. At $\rho = 4$, the PAR radius is 62.23% larger than the AM radius. Thus, the spatial relation between the two sublattice textures contributes to the equilibrium size of the composite skyrmion.

The interchanged strong and weak exchange directions also allow a common directional exchange modulation to

produce $\delta R \neq 0$ at $B_z = 0$. The slope $\chi_\epsilon = (d\delta R/d\delta)_0$ vanishes when the two sublattice exchange tensors are identical and increases over the sampled exchange contrasts. The perpendicular sublattice geometry therefore allows the relative radii to be changed by the same exchange modulation on both sublattices. This mechanism differs from the magnetic-field coupling, which remains nonzero in the isotropic reference.

Beyond these static deformations, the altermagnetic skyrmion supports an internal oscillation of δR coupled to the uniform helicity $\bar{\phi}$. A short perpendicular field pulse excites the two coordinates with a phase difference of approximately one quarter of a period. At $\rho = 2$, the LLG frequency is 49.52 GHz. Using P and κ_{con} from the same fixed- q family and $\kappa_{\bar{\phi}}$ from a common spin rotation gives 53.00 GHz. The predicted frequency and amplitude ratio are 7.03% and 9.58% above the LLG values, respectively. The relation $\omega_0 = \sqrt{\kappa_{\text{con}} \kappa_{\bar{\phi}}}/|P|$ describes this motion through the radial and helicity energy coefficients and their Berry coupling. In the LLG calculations, the AM frequency decreases to 46.51 GHz at $\rho = 4$. The altermagnetic skyrmion thus supports a field-excitabile internal mode whose frequency varies with the exchange contrast.

These results suggest a possible use of the internal mode in magnetic resonant elements operating at microwave frequencies. Calculations under an oscillating perpendicular field could determine the absorption spectrum and the amplitude of the oscillating magnetic moment, providing a basis for evaluating magnetic detection of the mode. Further work should assess thermal stability and establish whether lattice strain can tune the oscillation frequency through changes in the exchange coefficients, anisotropy, and DMI of a specific material.

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Supplemental Material for “Antisymmetric breathing in altermagnetic skyrmions”

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S1. MAGNETOSTATIC ENERGY AND FIELD

We calculate the magnetostatic contribution to the energy and effective fields in main-text Eqs. (2) and (3). The magnetization is uniform through the film thickness t and periodic in the film plane. On the spatial grid,

$$\mathbf{M}_{ij} = M_s(\mathbf{m}_{A,ij} + \mathbf{m}_{B,ij}). \quad (\text{S1})$$

For a grid field $\mathbf{U}$, we use the discrete Fourier pair

$$\begin{aligned} \hat{\mathbf{U}}_{n_x n_y} &= \sum_{i=0}^{N_x-1} \sum_{j=0}^{N_y-1} \mathbf{U}_{ij} e^{-2\pi i(n_x i/N_x + n_y j/N_y)}, \\ \mathbf{U}_{ij} &= \frac{1}{N_x N_y} \sum_{n_x, n_y} \hat{\mathbf{U}}_{n_x n_y} e^{2\pi i(n_x i/N_x + n_y j/N_y)}. \end{aligned}$$

Here N_x, N_y are the numbers of grid points, and the integer frequency indices are interpreted with the usual positive and negative Fourier wavevectors, $k_x = 2\pi n_x/(N_x h)$ and $k_y = 2\pi n_y/(N_y h)$. For $k = |\mathbf{k}|$, define

$$f_{\text{ms}}(kt) = \begin{cases} (1 - e^{-kt})/(kt), & k > 0, \\ 1, & k = 0. \end{cases} \quad (\text{S2})$$

The magnetostatic tensor is

$$\begin{aligned} N_{\alpha\beta}(\mathbf{k}) &= [1 - f_{\text{ms}}(kt)] \frac{k_\alpha k_\beta}{k^2}, \quad (k > 0), \\ N_{zz}(\mathbf{k}) &= f_{\text{ms}}(kt), \\ N_{xz} = N_{zx} = N_{yz} = N_{zy} &= 0, \\ N(\mathbf{0}) &= \text{diag}(0, 0, 1). \end{aligned} \quad (\text{S3})$$

Here $\alpha, \beta \in \{x, y\}$.

The field is

$$\mathbf{H}_{\text{ms}} = -\mu_0 \mathcal{F}^{-1}[N(\mathbf{k}) \mathcal{F} \mathbf{M}], \quad (\text{S4})$$

and its energy is

$$E_{\text{ms}} = -\frac{h^2 t}{2} \sum_{ij} \mathbf{M}_{ij} \cdot \mathbf{H}_{\text{ms},ij}. \quad (\text{S5})$$

For either sublattice,

$$-\frac{1}{M_s h^2 t} \frac{\partial E_{\text{ms}}}{\partial \mathbf{m}_{\eta,ij}} = \mathbf{H}_{\text{ms},ij}, \quad \eta = A, B. \quad (\text{S6})$$

Thus the same $\mathbf{H}_{\text{ms}}$, in tesla, enters both LLG equations.

For $\mathbf{m}_A(\mathbf{r}) = -\mathbf{m}_B(\mathbf{r})$ at every point,

$$\mathbf{M} \equiv 0, \quad \mathbf{H}_{\text{ms}} \equiv 0, \quad E_{\text{ms}} = 0.$$

The Fourier contribution from M_z contains both sublattices:

$$f_{\text{ms}} |\widehat{M}_z|^2 = M_s^2 f_{\text{ms}} (|\widehat{m}_{Az}|^2 + |\widehat{m}_{Bz}|^2 + 2 \text{Re}[\widehat{m}_{Az}^* \widehat{m}_{Bz}]).$$

The anisotropy coefficient remains $K = K_0/v_c$. Because $N_{xz} = N_{yz} = 0$, the two contributions to the energy are

$$E_{\text{ms}}^{\parallel} = -\frac{h^2 t}{2} \sum_{ij} \mathbf{M}_{\parallel,ij} \cdot \mathbf{H}_{\text{ms},\parallel,ij},$$

$$E_{\text{ms}}^z = -\frac{h^2 t}{2} \sum_{ij} M_{z,ij} H_{\text{ms},z,ij},$$

$$E_{\text{ms}} = E_{\text{ms}}^{\parallel} + E_{\text{ms}}^z.$$

For AM at $B_z = 0$ and $\rho = 2$,

$$E_{\text{ms}}^{\parallel} = 8.96 \times 10^{-22} \text{ J}, \quad E_{\text{ms}}^z = 2.18 \times 10^{-21} \text{ J}.$$

They contribute approximately 29% and 71% of E_{ms} .

S2. STATIC MINIMIZATION AND CONTOUR MEASUREMENTS

We minimize the total energy E of main-text Eq. (2). E is measured in joules and excludes all numerical constraint terms. The primary grid is

$$\begin{aligned} N_x = N_y &= 192, \quad h = 0.5 \text{ nm}, \\ N_x h = N_y h &= 96 \text{ nm}, \quad t = 1 \text{ nm}. \end{aligned}$$

The boundary conditions are periodic in x and y . We use

$$\begin{aligned} \bar{A} &= \sqrt{2} \times 10^{-11} \text{ J/m}, \quad A_0 = 2 \times 10^5 \text{ J/m}^3, \\ K &= 6 \times 10^5 \text{ J/m}^3, \quad M_s = 5.8 \times 10^5 \text{ A/m}, \\ D &= 2.5 \times 10^{-3} \text{ J/m}^2, \quad A_1 = \bar{A} \sqrt{\rho}, \quad A_2 = \bar{A} / \sqrt{\rho}. \end{aligned}$$

The primary static data comprise $\rho = 1, 1.5, 2, 2.5, 3, 3.5, 4$ at $B_z = 0, \pm 0.05 \text{ T}$ for AM and PAR. In AM, the exchange coefficients along (x, y) are (A_1, A_2) on A and (A_2, A_1) on B. In PAR, both sublattices use (A_1, A_2) .

For a grid function u_{ij} , define the central differences

$$D_x^c u_{ij} = \frac{u_{i+1,j} - u_{i-1,j}}{2h}, \quad D_y^c u_{ij} = \frac{u_{i,j+1} - u_{i,j-1}}{2h}.$$

The exchange energy uses nearest-neighbor spin differences, and the DMI uses D_x^c, D_y^c . Its negative energy-density gradient is

$$-\frac{1}{h^2 t} \frac{\partial E_D}{\partial \mathbf{m}_{\eta,ij}} = 2D \begin{pmatrix} D_x^c m_{\eta z} \\ D_y^c m_{\eta z} \\ -D_x^c m_{\eta x} - D_y^c m_{\eta y} \end{pmatrix}_{ij}. \quad (\text{S7})$$

Division by M_s gives the DMI field in tesla. The forward DMI bonds in main-text Eq. (1) have the continuum expansions

$$\begin{aligned} & -D_0(\mathbf{m} \times \mathbf{m}_{i+1,j}) \cdot \hat{\mathbf{e}}_y \\ & = -D_0 a(m_z \partial_x m_x - m_x \partial_x m_z) + O(a^2), \\ & + D_0(\mathbf{m} \times \mathbf{m}_{i,j+1}) \cdot \hat{\mathbf{e}}_x \\ & = -D_0 a(m_z \partial_y m_y - m_y \partial_y m_z) + O(a^2). \end{aligned}$$

Consequently $D = -D_0 a/v_c$. The initial Néel orientation is chosen by evaluating the two opposite radial orientations and selecting the one with the lower DMI energy. The magnetostatic energy and field of Sec. S1 are included at every evaluation.

The minimization uses Cartesian energy descent, limited-memory Broyden–Fletcher–Goldfarb–Shanno (LBFGS) minimization in angular coordinates, and a final Cartesian descent. For an unconstrained state,

$$\begin{aligned} \mathbf{g}_{\eta,ij} &= -\frac{1}{h^2 t} \frac{\partial E}{\partial \mathbf{m}_{\eta,ij}}, \\ \mathbf{g}_{\eta,ij}^\perp &= \mathbf{g}_{\eta,ij} - \mathbf{m}_{\eta,ij}(\mathbf{m}_{\eta,ij} \cdot \mathbf{g}_{\eta,ij}), \\ g_{\max} &= \max_{\eta,ij} |\mathbf{g}_{\eta,ij}^\perp|, \\ g_{\text{rms}} &= \left[\frac{1}{2N_x N_y} \sum_{\eta,ij} |\mathbf{g}_{\eta,ij}^\perp|^2 \right]^{1/2}. \end{aligned}$$

The acceptance conditions are

$$g_{\max} \leq 1 \text{ J/m}^3, \quad g_{\text{rms}} \leq 0.1 \text{ J/m}^3,$$

$$\max_{\ell \in \mathcal{C}} R_\eta^{(\ell)} - \min_{\ell \in \mathcal{C}} R_\eta^{(\ell)} < 10^{-4} \text{ nm}, \quad \eta = A, B.$$

The set $\mathcal{C}$ contains the last five recorded Cartesian states, recorded every 200 steps, and the final state.

The spatial origin is fixed at the center of the grid:

$$x_i = h \left(i - \frac{N_x - 1}{2} \right), \quad y_j = h \left(j - \frac{N_y - 1}{2} \right).$$

Along a ray at angle φ , the bilinearly interpolated values of $m_{\eta z}$ are sampled at $r_\ell = 0.10h\ell$. Let z_ℓ and $z_{\ell+1}$ bracket the first sign change along this ray. The contour crossing is obtained from

$$r_\eta(\varphi) = r_\ell - \frac{z_\ell}{z_{\ell+1} - z_\ell} (0.10h).$$

The measured axial distances are

$$\begin{aligned} b_\eta &= \frac{r_\eta(0) + r_\eta(\pi)}{2}, \\ a_\eta &= \frac{r_\eta(\pi/2) + r_\eta(3\pi/2)}{2}, \\ R_\eta &= \sqrt{a_\eta b_\eta}, \quad \varepsilon_\eta = \frac{b_\eta - a_\eta}{b_\eta + a_\eta}. \end{aligned}$$

These give the main-text coordinates

$$\begin{aligned} \bar{R} &= \frac{R_A + R_B}{2}, \quad \delta R = R_A - R_B, \\ e_s &= \frac{\varepsilon_A - \varepsilon_B}{2}, \quad e_u = \frac{\varepsilon_A + \varepsilon_B}{2}. \end{aligned}$$

The complete contour is sampled at $\varphi_j = 2\pi j/180$, $j = 0, \dots, 180$. The values at $j = 0$ and $j = 180$ refer to the same direction. Calculations requiring distinct directions use $j = 0, \dots, 179$.

S3. SYMMETRY UNDER $\mathcal{P}$

Magnetization transforms as an axial vector. Reflection about $x = y$ combined with sublattice exchange therefore acts as Eq. (S8).

$$\begin{aligned} \mathcal{P}: \quad \mathbf{m}'_A(x, y) &= -\mathcal{R} \mathbf{m}_B(y, x), \\ \mathbf{m}'_B(x, y) &= -\mathcal{R} \mathbf{m}_A(y, x). \end{aligned} \quad (\text{S8})$$

Here $\mathcal{R}(m_x, m_y, m_z) = (m_y, m_x, m_z)$. Primed fields are functions of (x, y) , while the unprimed fields on the right are evaluated at $(X, Y) = (y, x)$. The derivatives therefore transform by the chain rule as follows.

$$\begin{aligned} \partial_x [F(y, x)] &= [\partial_Y F](y, x), \\ \partial_y [F(y, x)] &= [\partial_X F](y, x). \end{aligned} \quad (\text{S9})$$

that is, $\partial_x \leftrightarrow \partial_Y$ and $\partial_y \leftrightarrow \partial_X$.

Exchange. Using Eqs. (S8) and (S9), and $|\mathcal{R}\mathbf{v}| = |\mathbf{v}|$,

$$\begin{aligned} A_1 |\partial_x \mathbf{m}'_A|^2 &= A_1 |\partial_Y \mathbf{m}_B|^2, \\ A_2 |\partial_y \mathbf{m}'_A|^2 &= A_2 |\partial_X \mathbf{m}_B|^2, \\ A_2 |\partial_x \mathbf{m}'_B|^2 &= A_2 |\partial_Y \mathbf{m}_A|^2, \\ A_1 |\partial_y \mathbf{m}'_B|^2 &= A_1 |\partial_X \mathbf{m}_A|^2. \end{aligned} \quad (\text{S10})$$

The four exchange terms reproduce the original sum because the coefficients A_1 and A_2 are interchanged between the sublattices.

Intersublattice coupling. $A_0 \mathbf{m}'_A \cdot \mathbf{m}'_B = A_0 (-\mathcal{R} \mathbf{m}_B) \cdot (-\mathcal{R} \mathbf{m}_A) = A_0 \mathbf{m}_B \cdot \mathbf{m}_A$, invariant, since $\mathcal{R}$ is orthogonal and the two sign changes cancel.

Anisotropy. $m'_{Az} = -m_{Bz}$ and $m'_{Bz} = -m_{Az}$. The squared components are interchanged, $m'^2_{Az} = m^2_{Bz}$ and $m'^2_{Bz} = m^2_{Az}$, so their sum is unchanged.

Dzyaloshinskii-Moriya interaction. Write $w_D = D[m_z(\partial_x m_x + \partial_y m_y) - (m_x \partial_x m_z + m_y \partial_y m_z)]$. Using Eqs. (S8) and (S9), and noting that $\mathcal{R}$ exchanges the in-plane components while the overall sign is squared in every term,

$$\begin{aligned} \partial_x m'_{Ax} &= \partial_x [-m_{By}(y, x)] \\ &= -[\partial_Y m_{By}](y, x), \end{aligned} \quad (\text{S11})$$

$$\begin{aligned} \partial_y m'_{Ay} &= \partial_y [-m_{Bx}(y, x)] \\ &= -[\partial_X m_{Bx}](y, x). \end{aligned} \quad (\text{S12})$$

so

$$\begin{aligned} m'_{Az}(\partial_x m'_{Ax} + \partial_y m'_{Ay}) &= [m_{Bz} \partial_Y m_{By} \\ &\quad + m_{Bz} \partial_X m_{Bx}](y, x) \quad (\text{S13}) \\ &= [m_{Bz} \nabla \cdot \mathbf{m}_{B\parallel}](y, x). \end{aligned}$$

and likewise

$$m'_{Ax} \partial_x m'_{Az} + m'_{Ay} \partial_y m'_{Az} = [(\mathbf{m}_{B\parallel} \cdot \nabla) m_{Bz}](y, x). \quad (\text{S14})$$

Equations (S11)–(S14) give

$$\begin{aligned} w_D[\mathbf{m}'_A](x, y) &= w_D[\mathbf{m}_B](y, x), \\ w_D[\mathbf{m}'_B](x, y) &= w_D[\mathbf{m}_A](y, x). \end{aligned}$$

Since $dx dy = dX dY$ for $(X, Y) = (y, x)$,

$$E_D[\mathbf{m}'_A] + E_D[\mathbf{m}'_B] = E_D[\mathbf{m}_A] + E_D[\mathbf{m}_B].$$

Zeeman energy. The transformed magnetization satisfies

$$M'_z(x, y) = -M_z(y, x), \quad \mu'_z = -\mu_z.$$

Therefore

$$\begin{aligned} E_Z[\mathcal{P}\mathbf{m}; B_z] &= -B_z(-\mu_z) = -E_Z[\mathbf{m}; B_z], \\ E_Z[\mathcal{P}\mathbf{m}; -B_z] &= -(-B_z)(-\mu_z) = E_Z[\mathbf{m}; B_z]. \end{aligned}$$

Here $\mathbf{m}$ denotes the pair $(\mathbf{m}_A, \mathbf{m}_B)$.

Magnetostatic energy. With $\mathcal{R}(x, y, z) = (y, x, z)$,

$$\widehat{\mathbf{M}}'(\mathbf{k}) = -\mathcal{R} \widehat{\mathbf{M}}(\mathcal{R}\mathbf{k}), \quad N(\mathcal{R}\mathbf{k}) = \mathcal{R} N(\mathbf{k}) \mathcal{R}^T.$$

Consequently,

$$\widehat{\mathbf{M}}'(\mathbf{k})^\dagger N(\mathbf{k}) \widehat{\mathbf{M}}'(\mathbf{k}) = \widehat{\mathbf{M}}(\mathcal{R}\mathbf{k})^\dagger N(\mathcal{R}\mathbf{k}) \widehat{\mathbf{M}}(\mathcal{R}\mathbf{k}).$$

The sum over wavevectors in the square periodic cell gives $E_{\text{ms}}[\mathcal{P}\mathbf{m}] = E_{\text{ms}}[\mathbf{m}]$. Here $\dagger$ denotes complex conjugate transpose.

Collective coordinates. The contour transforms as

$$\begin{aligned} r'_A(\varphi) &= r_B(\pi/2 - \varphi), \quad r'_B(\varphi) = r_A(\pi/2 - \varphi), \\ a'_A &= b_B, \quad b'_A = a_B, \\ a'_B &= b_A, \quad b'_B = a_A. \end{aligned} \quad (\text{S15})$$

Thus $R'_A = R_B$, $R'_B = R_A$, $\varepsilon'_A = -\varepsilon_B$, and $\varepsilon'_B = -\varepsilon_A$, giving

$$(\bar{R}, e_s, \delta R, e_u) \mapsto (\bar{R}, e_s, -\delta R, -e_u). \quad (\text{S16})$$

Combining this result with the field reversal gives

$$E(\bar{R}, e_s, \delta R, e_u; B_z) = E(\bar{R}, e_s, -\delta R, -e_u; -B_z).$$

For a term $\delta R^{n_R} e_u^{n_e} B_z^{n_B}$ in the expansion at fixed $\bar{R}, e_s$, with nonnegative integer exponents,

$$n_R + n_e + n_B \text{ must be even.}$$

In particular, $B_z \delta R$, $B_z e_u$, and $\delta R e_u$ are allowed. For the smooth equilibrium branch connected to the zero-field state $\delta R = e_u = 0$,

$$\begin{aligned} \delta R(-B_z) &= -\delta R(B_z), \quad e_u(-B_z) = -e_u(B_z), \\ \bar{R}(-B_z) &= \bar{R}(B_z), \quad e_s(-B_z) = e_s(B_z). \end{aligned}$$

Section S7 estimates the coefficient Λ_{e_u} of the field coupling to e_u .

A rotation by $\pi/2$ about z , combined with sublattice exchange, acts as

$$\begin{aligned} \mathbf{m}'_A(\mathbf{r}) &= R_z(\pi/2) \mathbf{m}_B(R_z(-\pi/2)\mathbf{r}), \\ \mathbf{m}'_B(\mathbf{r}) &= R_z(\pi/2) \mathbf{m}_A(R_z(-\pi/2)\mathbf{r}). \end{aligned}$$

Here R_z rotates both position and spin vectors about z . This transformation gives

$$\begin{aligned} (m_{Az}^{\text{core}}, m_{Bz}^{\text{core}}) &: (+1, -1) \mapsto (-1, +1), \\ (\delta R, e_u, \mu_z, B_z) &\mapsto (-\delta R, -e_u, \mu_z, B_z). \end{aligned}$$

The energy is unchanged at the same B_z , but the core polarities are reversed. The field-parity relations above apply to the fixed core polarities $(+1, -1)$ used in the calculations.

For the quadratic energy in main-text Eq. (8), stationarity at fixed $\bar{R}$ and e_s gives

$$\begin{pmatrix} \kappa_0 & \zeta \\ \zeta & \kappa_{e_u} \end{pmatrix} \begin{pmatrix} \delta R \\ e_u \end{pmatrix} = B_z \begin{pmatrix} \Lambda_R \\ \Lambda_{e_u} \end{pmatrix}.$$

For $\kappa_{e_u} > 0$, eliminating e_u gives

$$\begin{aligned} \kappa &= \kappa_0 - \frac{\zeta^2}{\kappa_{e_u}}, \\ e_u &= \frac{\Lambda_{e_u} B_z - \zeta \delta R}{\kappa_{e_u}}, \\ \delta R &= \frac{\Lambda_R - \zeta \Lambda_{e_u} / \kappa_{e_u}}{\kappa} B_z \equiv \chi B_z. \end{aligned}$$

At $B_z = 0$, with δR prescribed,

$$e_u = -\frac{\zeta}{\kappa_{e_u}} \delta R, \quad \Delta E(\delta R) = \frac{1}{2} \kappa \delta R^2 + O(\delta R^4),$$

where ΔE is the energy relative to $\delta R = 0$ after minimizing e_u at the same $\bar{R}, e_s$. The quadratic form is positive when

$$\kappa_{e_u} > 0, \quad \kappa_0 \kappa_{e_u} - \zeta^2 > 0,$$

which is equivalent to $\kappa_{e_u} > 0$ and $\kappa > 0$.

S4. SMALL-FIELD SLOPES AND NUMERICAL CHECKS

The zero-field slope in main-text Eq. (10) is estimated from independently minimized states at $B_z = \pm B_1$. We

calculate

$$\mu_z(B_z) = M_s t h^2 \sum_{ij} [m_{Az,ij}(B_z) + m_{Bz,ij}(B_z)],$$

$$\Delta X = X(B_1) - X(-B_1).$$

For $B_1 = 0.05$ T,

$$\chi = \left. \frac{d\delta R}{dB_z} \right|_0 \simeq \frac{\Delta\delta R}{2B_1}, \quad (\text{S17})$$

$$\Lambda_{\text{path}} = \frac{\Delta\mu_z}{\Delta\delta R}.$$

For an analytic odd dependence $\delta R(B_z) = \chi B_z + c_3 B_z^3 + O(B_z^5)$,

$$\frac{\Delta\delta R}{2B_1} = \chi + c_3 B_1^2 + O(B_1^4).$$

Thus the difference at a specified B_1 estimates the derivative at zero. All spin variables can change between the two states defining Λ_{path} . For the $h = 0.5$ nm grid,

ρ	geometry	χ (nm/T)	Λ_{path} (J T ⁻¹ m ⁻¹)
2	AM	3.08	3.73×10^{-11}
4	AM	3.49	3.50×10^{-11}
4	PAR	3.70	4.89×10^{-11}

The same differences are evaluated for the seven ρ values and both geometries specified in Sec. S2. The zero-field states give $\bar{R}$ in main-text Table I. The states at ± 0.05 T give the listed estimates of χ .

At $\rho = 2$, define the even difference

$$\Delta e_s^{\text{even}}(B_1) = \frac{e_s(B_1) + e_s(-B_1)}{2} - e_s(0).$$

The additional small-field calculations give

B_1 (T)	Δe_s^{even}	$\Delta e_s^{\text{even}}/B_1^2$ (T ⁻²)
0.0125	2.79×10^{-6}	1.79×10^{-2}
0.025	-7.23×10^{-5}	-1.16×10^{-1}

For $\rho = 2$ in the same 96×96 nm² periodic cell, using $B_z = \pm 0.05$ T,

h (nm)	$ \chi_{\parallel} - \chi_{\text{AM}} $ (nm/T)
0.50	1.83×10^{-2}
0.25	6.17×10^{-2}

PAR has the larger estimate of χ on both grids, while the magnitude of the difference remains sensitive to h .

S5. CONTOUR EXPANSION AND LOCAL MAGNETIZATION

For the elliptical contour of sublattice η ,

$$\frac{x^2}{b_\eta^2} + \frac{y^2}{a_\eta^2} = 1, \quad x = r \cos \varphi, \quad y = r \sin \varphi.$$

Its polar radius is

$$r_\eta(\varphi) = \frac{a_\eta b_\eta}{\sqrt{a_\eta^2 \cos^2 \varphi + b_\eta^2 \sin^2 \varphi}}. \quad (\text{S18})$$

Using the main-text definitions $R_\eta = \sqrt{a_\eta b_\eta}$ and $\varepsilon_\eta = (b_\eta - a_\eta)/(b_\eta + a_\eta)$ gives

$$a_\eta = R_\eta \sqrt{\frac{1 - \varepsilon_\eta}{1 + \varepsilon_\eta}}, \quad b_\eta = R_\eta \sqrt{\frac{1 + \varepsilon_\eta}{1 - \varepsilon_\eta}}, \quad (\text{S19})$$

$$\frac{r_\eta(\varphi)}{R_\eta} = \frac{\sqrt{1 - \varepsilon_\eta^2}}{\sqrt{1 - 2\varepsilon_\eta \cos 2\varphi + \varepsilon_\eta^2}}. \quad (\text{S20})$$

Expanding in ε_η ,

$$\begin{aligned} \frac{r_\eta}{R_\eta} &= 1 + \varepsilon_\eta \cos 2\varphi - \varepsilon_\eta^2 + \frac{3}{2} \varepsilon_\eta^2 \cos^2 2\varphi + O(\varepsilon_\eta^3) \\ &= 1 + \varepsilon_\eta \cos 2\varphi - \frac{\varepsilon_\eta^2}{4} + \frac{3\varepsilon_\eta^2}{4} \cos 4\varphi + O(\varepsilon_\eta^3). \end{aligned} \quad (\text{S21})$$

The exact ellipse obeys

$$\frac{1}{2} \int_0^{2\pi} r_\eta^2 d\varphi = \pi a_\eta b_\eta = \pi R_\eta^2.$$

In the expansion, the mean contribution from the terms of order ε_η^2 in r_η^2/R_η^2 is

$$\left\langle -\frac{\varepsilon_\eta^2}{2} + \varepsilon_\eta^2 \cos^2 2\varphi + \frac{3}{2} \varepsilon_\eta^2 \cos 4\varphi \right\rangle_\varphi = 0,$$

where $\langle X \rangle_\varphi = (2\pi)^{-1} \int_0^{2\pi} X d\varphi$. For the numerical contours, R_η is measured from the axial crossings as specified in Sec. S2.

For a centered contour symmetric about the x and y axes,

$$\frac{r_\eta(\varphi)}{R_\eta} = c_{0,\eta} + \sum_{n \geq 1} c_{2n,\eta} \cos(2n\varphi),$$

$$c_{0,\eta} = \frac{1}{2\pi} \int_0^{2\pi} \frac{r_\eta(\varphi)}{R_\eta} d\varphi,$$

$$c_{2n,\eta} = \frac{1}{\pi} \int_0^{2\pi} \frac{r_\eta(\varphi)}{R_\eta} \cos(2n\varphi) d\varphi.$$

For an ellipse,

$$\left| \frac{c_{4,\eta}}{c_{2,\eta}} \right| = \frac{3}{4} |\varepsilon_\eta| + O(\varepsilon_\eta^3), \quad \left| \frac{c_{6,\eta}}{c_{2,\eta}} \right| = \frac{5}{8} \varepsilon_\eta^2 + O(\varepsilon_\eta^4).$$

For the zero-field AM state invariant under $\mathcal{P}$,

$$R_A = R_B = \bar{R}, \quad c_{2n,B} = (-1)^n c_{2n,A},$$

$$\frac{u(\varphi)}{\bar{R}} = 2 \sum_{j \geq 0} c_{4j+2,A} \cos[(4j+2)\varphi].$$

Thus the common $\cos 4\varphi$ term cancels, whereas $\cos 2\varphi$ and $\cos 6\varphi$ are allowed in u .

For the approximation used in main-text Eq. (11),

$$\begin{aligned} m_{Az}(r, \varphi) &= f\left(\frac{r}{r_A(\varphi)}\right), \\ m_{Bz}(r, \varphi) &= -f\left(\frac{r}{r_B(\varphi)}\right), \end{aligned} \quad (\text{S22})$$

where $f(0) = 1$, $f(1) = 0$, and $f(\infty) = -1$. At fixed r ,

$$\left. \frac{\partial f(r/\ell)}{\partial \ell} \right|_{\ell=\bar{R}} = -\frac{r}{\bar{R}^2} f'\left(\frac{r}{\bar{R}}\right).$$

To first order in $r_A(\varphi) - \bar{R}$ and $r_B(\varphi) - \bar{R}$,

$$\begin{aligned} M_z(r, \varphi) &= M_s \left[f\left(\frac{r}{r_A}\right) - f\left(\frac{r}{r_B}\right) \right] \\ &\simeq M_s p(r) u(\varphi), \\ p(r) &= -\frac{r}{\bar{R}^2} f'\left(\frac{r}{\bar{R}}\right). \end{aligned} \quad (\text{S23})$$

The function p has units of inverse length. It is positive where f decreases through the wall. Separately, the first-order expansion of each elliptical contour gives

$$\begin{aligned} u(\varphi) &= r_A(\varphi) - r_B(\varphi) \\ &\simeq \delta R + (R_A \varepsilon_A - R_B \varepsilon_B) \cos 2\varphi \\ &= \delta R + (2\bar{R} e_s + \delta R e_u) \cos 2\varphi. \end{aligned} \quad (\text{S24})$$

At zero field, $\delta R = e_u = 0$, so

$$u(\varphi) \simeq 2\bar{R} e_s \cos 2\varphi.$$

For the calculated $\rho = 2$ state,

$$\begin{aligned} u(\varphi) &\simeq (1.29 \text{ nm}) \cos 2\varphi, \\ \frac{2\bar{R} e_s}{\lambda} &= 2.68 \times 10^{-1}, \quad \lambda = \sqrt{\bar{A}/K}, \\ \frac{\max_{\mathbf{r}} |M_z(\mathbf{r})|}{M_s} &= 4.65 \times 10^{-1}, \\ \frac{\max_{\mathbf{r}} |M_z(\mathbf{r})|/M_s}{2\bar{R} e_s/\lambda} &= 1.73. \end{aligned}$$

The numerical moments in Sec. S6 are evaluated from the complete spin configurations, rather than from the first-order expression for M_z .

S6. NET MOMENT AND HIGHER MAGNETIC MOMENTS

The moments used below are

$$\mu_z = t \int d^2 r M_z, \quad \mathcal{M}_z^{(3)} = t \int d^2 r (x^2 - y^2) M_z. \quad (\text{S25})$$

Here $x^2 - y^2 = r^2 \cos 2\varphi$, $[\mu_z] = \text{A m}^2 = \text{J/T}$, and $[\mathcal{M}_z^{(3)}] = \text{A m}^4$. The superscript 3 counts one magnetization component and two position factors in this Cartesian moment [1, 2].

Assume the radial moments converge and define

$$W_n = \int_0^\infty r^{1+n} p(r) dr, \quad [W_n] = \text{m}^{n+1}.$$

Using $\int_0^{2\pi} \cos 2\varphi d\varphi = 0$ in the first-order magnetization gives

$$\mu_z = 2\pi M_s t W_0 \delta R \equiv \Lambda_R \delta R. \quad (\text{S26})$$

At $\delta R = e_u = 0$, retaining the leading term in e_s and using $\int_0^{2\pi} \cos^2 2\varphi d\varphi = \pi$ gives

$$\mathcal{M}_z^{(3)} \simeq 2\pi M_s t W_2 \bar{R} e_s. \quad (\text{S27})$$

For an area estimate, replace the smooth radial function by

$$f(\xi) = \begin{cases} +1, & 0 \leq \xi < 1, \\ -1, & \xi > 1. \end{cases}$$

$f'(\xi) = -2\delta_D(\xi - 1)$, where δ_D is the Dirac delta distribution. Hence

$$p(r) = \frac{2r}{\bar{R}} \delta_D(r - \bar{R}), \quad W_0 = 2\bar{R}, \quad W_2 = 2\bar{R}^3.$$

For exact elliptical interiors $\mathcal{D}_A, \mathcal{D}_B$,

$$\begin{aligned} \mu_z &= 2M_s t \left(\int_{\mathcal{D}_A} d^2 r - \int_{\mathcal{D}_B} d^2 r \right) \\ &= 2\pi M_s t (R_A^2 - R_B^2) = 4\pi M_s t \bar{R} \delta R. \end{aligned}$$

This gives $\Lambda_R = 4\pi M_s t \bar{R}$ in main-text Eq. (16). For the same piecewise profile,

$$\begin{aligned} \mathcal{M}_z^{(3)} &= \frac{\pi M_s t}{2} [R_A^2 (b_A^2 - a_A^2) - R_B^2 (b_B^2 - a_B^2)] \\ &= 2\pi M_s t \left[\frac{R_A^4 \varepsilon_A}{1 - \varepsilon_A^2} - \frac{R_B^4 \varepsilon_B}{1 - \varepsilon_B^2} \right]. \end{aligned}$$

We express deviations from the leading area estimates by

$$\Lambda_R = 4\pi c_\mu M_s t \bar{R}, \quad \mathcal{M}_z^{(3)} = 4\pi c_Q M_s t \bar{R}^4 e_s. \quad (\text{S28})$$

The second definition is used for zero-field states with $e_s \neq 0$. For the piecewise profile, $c_\mu = 1$. At $\delta R = e_u = 0$,

$$\mathcal{M}_z^{(3)} = 4\pi M_s t \bar{R}^4 \frac{e_s}{1 - e_s^2} = 4\pi M_s t \bar{R}^4 e_s + O(e_s^3), \quad (\text{S29})$$

so $c_Q = (1 - e_s^2)^{-1}$ for exact ellipses, and $c_Q = 1$ to leading order in e_s .

$$\left. \frac{\partial^2 \mathcal{M}_z^{(3)}}{\partial \delta R \partial e_u} \right|_{\delta R = e_u = 0} = 8\pi M_s t \bar{R}^3 \frac{1 + e_s^2}{(1 - e_s^2)^2},$$

with $\bar{R}, e_s$ fixed. Its leading coefficient at small e_s is $8\pi M_s t \bar{R}^3$, multiplying $\delta R e_u$.

For each computed zero-field state, we evaluate

$$c_Q = \frac{\mathcal{M}_z^{(3)}}{4\pi M_s t \bar{R}^4 e_s}.$$

The values are

ρ	1.5	2	2.5	3	3.5	4
c_Q	5.53	5.94	6.39	6.80	7.15	7.46

The radial weight in $\mathcal{M}_z^{(3)}$ contains two more powers of r than that in μ_z . Magnetization outside the central part of the wall therefore contributes more strongly to c_Q . Both c_Q and $\bar{R}$ change along this scan. At $\rho = 1$, $\mathcal{M}_z^{(3)} = 0$. For AM at $\rho = 2$,

$$\frac{\Lambda_{\text{path}}}{4\pi M_s t \bar{R}} = 1.32,$$

$$\Lambda_{\text{path}} = \frac{\mu_z(B_1) - \mu_z(-B_1)}{\delta R(B_1) - \delta R(-B_1)}, \quad B_1 = 0.05 \text{ T}.$$

This ratio uses the field-minimized states of Sec. S4.

For an exact ellipse and any common scaled function f , use $x = b_\eta \xi \cos \vartheta$ and $y = a_\eta \xi \sin \vartheta$. Then

$$\frac{r}{r_\eta(\varphi)} = \xi, \quad d^2 r = a_\eta b_\eta \xi d\xi d\vartheta,$$

$$\int d^2 r [1 + s_\eta m_{\eta z}] = 2\pi R_\eta^2 \int_0^\infty \xi [1 + f(\xi)] d\xi,$$

$$s_A = +1, \quad s_B = -1.$$

Consequently,

$$\mu_z = 2\pi M_s t (R_A^2 - R_B^2) \int_0^\infty \xi [1 + f(\xi)] d\xi,$$

$$\left. \frac{\partial \mu_z}{\partial e_u} \right|_{R_A, R_B, f} = 0.$$

Thus $\Lambda_{e_u} = 0$ within this specified family, as used in main-text Eq. (17). The numerical moment instead uses the complete spin configuration:

$$\mathcal{M}_z^{(3)} = M_s t h^2 \sum_{ij} (m_{Az,ij} + m_{Bz,ij}) (x_i^2 - y_j^2). \quad (\text{S30})$$

For a radial trial state, write

$$\mathbf{m}_{A\parallel} = f_\parallel(r/r_A) \hat{\mathbf{r}}, \quad \mathbf{m}_{B\parallel} = -f_\parallel(r/r_B) \hat{\mathbf{r}},$$

where $f_\parallel$ is the signed radial component of the reference unit spin and $f_\parallel^2 + f^2 = 1$. To first order in the contour displacements,

$$\mathbf{M}_\parallel \simeq \tilde{p}(r) u(\varphi) \hat{\mathbf{r}},$$

$$\tilde{p}(r) = -\frac{M_s r}{\bar{R}^2} f'_\parallel\left(\frac{r}{\bar{R}}\right), \quad [\tilde{p}] = \text{A/m}^2.$$

The position-weighted in-plane moment is

$$\mathcal{M}_{ij}^{(2)} = t \int d^2 r r_i M_j, \quad i, j \in \{x, y\}, \quad (\text{S31})$$

$$[\mathcal{M}_{ij}^{(2)}] = \text{A m}^3.$$

At zero field, $u \simeq 2\bar{R}e_s \cos 2\varphi$. The angular integrals give

$$t \int d^2 r \mathbf{M}_\parallel = 0,$$

$$\begin{pmatrix} \mathcal{M}_{xx}^{(2)} & \mathcal{M}_{xy}^{(2)} \\ \mathcal{M}_{yx}^{(2)} & \mathcal{M}_{yy}^{(2)} \end{pmatrix} \simeq \pi t \bar{R} e_s \int_0^\infty r^2 \tilde{p}(r) dr \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Direct integration of the computed spin configurations gives

ρ	$-\mathcal{M}_{xx}^{(2)} (10^{-28} \text{ A m}^3)$
1.5	1.09
2	1.82
2.5	2.34
3	2.76
3.5	3.10
4	3.41

with $\mathcal{M}_{yy}^{(2)} = -\mathcal{M}_{xx}^{(2)}$ to the reported precision. The ratio to the contour-displacement scale is

$$\frac{-\mathcal{M}_{xx}^{(2)}}{2\bar{R}e_s/\lambda} = \begin{cases} 6.81 \times 10^{-28} \text{ A m}^3, & \rho = 1.5, \\ 6.87 \times 10^{-28} \text{ A m}^3, & \rho = 4. \end{cases} \quad (\text{S32})$$

The endpoint values differ by approximately 1.0%. At $\rho = 1$, $\mathbf{M} \equiv 0$ and $\mathcal{M}_{ij}^{(2)} = 0$. For a field-dependent state, the sum of the diagonal components is

$$\mathcal{M}_{xx}^{(2)} + \mathcal{M}_{yy}^{(2)} = t \int d^2 r (x M_x + y M_y),$$

$$[\mathcal{M}_{xx}^{(2)} + \mathcal{M}_{yy}^{(2)}](-B_z) = -[\mathcal{M}_{xx}^{(2)} + \mathcal{M}_{yy}^{(2)}](B_z).$$

For an isolated skyrmion and observation distance d much larger than its spatial extent, the rank-2 and rank-3 field contributions have radial dependences d^{-4} and d^{-5} , respectively.

S7. FIELD COUPLING TO THE UNIFORM ELLIPTICITY

For AM at $B_z = 0$ and $\rho = 2$, we vary the integral-based ellipticity constraint and adjust the area constraint so that the measured radius difference remains near zero. The constrained quantities are

$$q = \sqrt{S_A/\pi} - \sqrt{S_B/\pi},$$

$$\tilde{e}_u = \frac{\tilde{\varepsilon}_A + \tilde{\varepsilon}_B}{2}, \quad \tilde{\varepsilon}_\eta = \frac{\pi T_\eta}{S_\eta^2}. \quad (\text{S33})$$

Here

$$\begin{aligned} S_\eta &= \frac{h^2}{2} \sum_{ij} (1 + s_\eta m_{\eta z, ij}), \\ T_\eta &= \frac{h^2}{2} \sum_{ij} (x_i^2 - y_j^2) (1 + s_\eta m_{\eta z, ij}), \\ s_A &= +1, \quad s_B = -1. \end{aligned}$$

The imposed targets and acceptance conditions are

$$\tilde{e}_u^* \in \{0, \pm 0.01, \pm 0.02\},$$

$$\begin{aligned} |q - q^*| &\leq 0.002 \text{ nm}, \\ |\tilde{e}_u - \tilde{e}_u^*| &\leq 2 \times 10^{-5}, \\ |\delta R| &\leq 0.002 \text{ nm}. \end{aligned}$$

The five measured values used in the fits are

$$e_u \simeq 0, \quad \pm 8.92 \times 10^{-4}, \quad \pm 1.77 \times 10^{-3}.$$

We fit μ_z and the physical energy against these measured values of e_u . The resulting coefficients estimate the fixed-coordinate derivatives in main-text Eq. (8).

The coefficients in main-text Eq. (8) are defined at $B_z = \delta R = e_u = 0$, with $\tilde{R}, e_s$ fixed:

$$\Lambda_{e_u} = \left. \frac{\partial \mu_z}{\partial e_u} \right|_0, \quad \kappa_{e_u} = \left. \frac{\partial^2 E}{\partial e_u^2} \right|_0.$$

Let e_j denote the measured e_u in the five calibrated states. The unweighted fits are

$$(\mu_{\text{off}}, \ell_\mu) = \underset{c, \ell}{\operatorname{argmin}} \sum_{j=1}^5 (\mu_{z,j} - c - \ell e_j)^2,$$

$$(E_{\text{off}}, b_E, k_e) = \underset{c, b, k}{\operatorname{argmin}} \sum_{j=1}^5 \left(E_j - c - b e_j - \frac{k}{2} e_j^2 \right)^2.$$

The fit coefficients estimate the main-text derivatives as $\Lambda_{e_u} \simeq \ell_\mu$ and $\kappa_{e_u} \simeq k_e$. The calibrated-state fits give the estimates

$$\Lambda_{e_u} \simeq 1.00809327 \times 10^{-19} \text{ J/T}, \quad (\text{S34})$$

$$\kappa_{e_u} \simeq 1.59355329 \times 10^{-17} \text{ J}. \quad (\text{S35})$$

Minimizing the quadratic energy in main-text Eq. (8) at prescribed δR gives

$$0 = \kappa_{e_u} e_u + \zeta \delta R - \Lambda_{e_u} B_z, \quad e_u = \frac{\Lambda_{e_u} B_z - \zeta \delta R}{\kappa_{e_u}}. \quad (\text{S36})$$

At unconstrained small-field equilibrium, $\delta R = \chi B_z + O(B_z^3)$, and therefore

$$e_u = \frac{\Lambda_{e_u} - \zeta \chi}{\kappa_{e_u}} B_z + O(B_z^3).$$

For the field pair $B_z = \pm 0.05 \text{ T}$,

$$\begin{aligned} \frac{\Delta e_u}{\Delta B_z} &= -4.42889 \times 10^{-3} \text{ T}^{-1}, \\ \frac{\Lambda_{e_u}}{\kappa_{e_u}} &= 6.32607 \times 10^{-3} \text{ T}^{-1}. \end{aligned}$$

Using the quadratic model to compare these values gives

$$-\frac{\zeta}{\kappa_{e_u}} \frac{\Delta \delta R}{\Delta B_z} \simeq \frac{\Delta e_u}{\Delta B_z} - \frac{\Lambda_{e_u}}{\kappa_{e_u}} \simeq -1.08 \times 10^{-2} \text{ T}^{-1},$$

$$\frac{\zeta}{\kappa_{e_u}} \simeq \frac{\Lambda_{e_u}/\kappa_{e_u} - \Delta e_u/\Delta B_z}{\Delta \delta R/\Delta B_z} \simeq 3.49 \times 10^{-3} \text{ nm}^{-1}.$$

This estimate uses the specified field interval and the calibrated-state coefficients above. The same field pair gives

$$\frac{\Delta e_u}{\Delta \delta R} = -1.43893 \times 10^{-3} \text{ nm}^{-1}, \quad e_s(0) = 0.16800.$$

Section S15 compares this interval-based estimate with e_u measured along the fixed- q states and during the LLG oscillation.

S8. RADIUS DIFFERENCE UNDER EXCHANGE MODULATION

We derive the local estimate used in main-text Eq. (19). Let $\mathbf{m}_0(\boldsymbol{\xi})$ be a fixed circular Néel reference profile, and set

$$\begin{aligned} \mathbf{m}_\eta(x, y) &= s_\eta \mathbf{m}_0(\boldsymbol{\xi}), \\ \boldsymbol{\xi} &= (x/b_\eta, y/a_\eta), \quad s_A = +1, \quad s_B = -1. \end{aligned}$$

This estimate retains the exchange, anisotropy, and DMI of one sublattice. All energies below include the film thickness t and are measured in joules. Define

$$\begin{aligned} I_{\text{ex}} &= \int d^2 \xi |\partial_{\xi_x} \mathbf{m}_0|^2 = \int d^2 \xi |\partial_{\xi_y} \mathbf{m}_0|^2, \\ I_K &= \int d^2 \xi (1 - m_{0z}^2), \\ I_D &= - \int d^2 \xi (m_{0z} \partial_{\xi_x} m_{0x} - m_{0x} \partial_{\xi_x} m_{0z}). \end{aligned}$$

The corresponding y integral also equals $-I_D$. The reference Néel orientation is chosen so that $DI_D > 0$. Using $dx dy = a_\eta b_\eta d^2 \xi$ gives

$$\begin{aligned} E_{\text{ex}, \eta} &= t I_{\text{ex}} \left(A_{\eta x} \frac{a_\eta}{b_\eta} + A_{\eta y} \frac{b_\eta}{a_\eta} \right), \\ E_{K, \eta} &= t K I_K a_\eta b_\eta, \\ E_{D, \eta} &= -t D I_D (a_\eta + b_\eta). \end{aligned} \quad (\text{S37})$$

Let $\tau_\eta = b_\eta/a_\eta$ and $R_\eta = \sqrt{a_\eta b_\eta}$. Then

$$E_\eta^{\text{loc}} = t \left[I_{\text{ex}} \left(\frac{A_{\eta x}}{\tau_\eta} + A_{\eta y} \tau_\eta \right) + K I_K R_\eta^2 - D I_D R_\eta g(\tau_\eta) \right], \quad (\text{S38})$$

$$g(\tau) = \sqrt{\tau} + \frac{1}{\sqrt{\tau}}.$$

Minimizing only the exchange term at $\delta = 0$ gives

$$\tau_A = \sqrt{\rho}, \quad \tau_B = \rho^{-1/2}, \quad E_{\text{ex},\eta}^{\text{min}} = 2t I_{\text{ex}} \bar{A}. \quad (\text{S39})$$

At prescribed τ_η , the radius minimizes

$$\frac{\partial E_\eta^{\text{loc}}}{\partial R_\eta} = t[2K I_K R_\eta - D I_D g(\tau_\eta)] = 0, \quad (\text{S40})$$

$$R_\eta = R_0 g(\tau_\eta), \quad R_0 = \frac{D I_D}{2K I_K}.$$

The radius minimum exists for $K I_K > 0$ and $D I_D > 0$. Since $g(\tau) = g(1/\tau)$, reciprocal aspect ratios give equal radii within this trial family. The numerical radii instead minimize the full energy, including intersublattice exchange and magnetostatics.

At $B_z = 0$, apply the same relative modulation to both sublattices:

$$(A_{Ax}, A_{Ay}) = (A_1(1 + \delta), A_2(1 - \delta)), \quad (\text{S41})$$

$$(A_{Bx}, A_{By}) = (A_2(1 + \delta), A_1(1 - \delta)).$$

Thus $\Delta A_{\eta x}/A_{\eta x}^{(0)} = \delta$ and $\Delta A_{\eta y}/A_{\eta y}^{(0)} = -\delta$. The conversion from δ to lattice strain requires material-specific exchange coefficients. The geometric means are

$$\sqrt{A_{\eta x} A_{\eta y}} = \bar{A} \sqrt{1 - \delta^2} = \bar{A} [1 - \delta^2/2 + O(\delta^4)].$$

The exchange-only aspect-ratio minima become

$$\tau_A(\delta) = \sqrt{\rho} \sqrt{\frac{1 + \delta}{1 - \delta}} = \sqrt{\rho}(1 + \delta) + O(\delta^2), \quad (\text{S42})$$

$$\tau_B(\delta) = \rho^{-1/2} \sqrt{\frac{1 + \delta}{1 - \delta}} = \rho^{-1/2}(1 + \delta) + O(\delta^2).$$

Inserting these into the local radius estimate gives

$$\delta R = R_0 [g(\tau_A) - g(\tau_B)]$$

$$= R_0 (\rho^{1/4} - \rho^{-1/4}) \left[\left(\frac{1 + \delta}{1 - \delta} \right)^{1/4} - \left(\frac{1 - \delta}{1 + \delta} \right)^{1/4} \right]. \quad (\text{S43})$$

For small δ ,

$$\delta R \simeq R_0 (\rho^{1/4} - \rho^{-1/4}) \delta. \quad (\text{S44})$$

The separate radius derivatives in this approximation are

$$\left. \frac{dR_A}{d\delta} \right|_0 = - \left. \frac{dR_B}{d\delta} \right|_0 = \frac{R_0}{2} (\rho^{1/4} - \rho^{-1/4}).$$

They have opposite signs for $\rho > 1$ and vanish at $\rho = 1$.

The full-energy calculations use

$$\delta \in \{0, \pm 0.02, \pm 0.04, \pm 0.06\}, \quad B_z = 0,$$

with K, D and the other unmodulated parameters fixed. Near $\delta = 0$,

$$\delta R(\delta) \simeq c_\epsilon + \chi_\epsilon \delta, \quad \chi_\epsilon = \left. \frac{d\delta R}{d\delta} \right|_0,$$

where the numerical slope is estimated by fitting the computed states with $|\delta| \leq 0.02$. The fitted values are

ρ	χ_ϵ (nm)
1	1.30×10^{-11}
1.5	3.15×10^{-11}
2	3.85×10^{-11}
2.5	6.48×10^{-11}
3	8.29×10^{-11}
3.5	1.09
4	1.19

At $\rho = 1$, symmetry gives $\chi_\epsilon = 0$. The numerical fit gives $c_\epsilon = 3.82 \times 10^{-8}$ nm and the residual slope shown above.

S9. LOCAL INTERSUBLATTICE ENERGY OF SEPARATED CONTOURS

The excess intersublattice energy is

$$E_{\text{inter}} = A_0 t \int d^2 r [1 + \mathbf{m}_A \cdot \mathbf{m}_B].$$

For a local estimate, neglect the wall curvature across each cross-section, replace the normal separation by $u(\varphi)$, and use the line element $ds \simeq \bar{R} d\varphi$. The two walls have a common width λ . Then

$$E_{\text{inter}} \simeq A_0 t \lambda \int_0^{2\pi} \bar{R} F\left(\frac{u(\varphi)}{\lambda}\right) d\varphi. \quad (\text{S45})$$

Let ℓ be the normal coordinate and define

$$\mathbf{n}_0(\xi) = (\text{sech } \xi, 0, \tanh \xi),$$

$$\mathbf{m}_A(\ell) = \mathbf{n}_0\left(\frac{\ell - u/2}{\lambda}\right),$$

$$\mathbf{m}_B(\ell) = -\mathbf{n}_0\left(\frac{\ell + u/2}{\lambda}\right).$$

With $\xi = \ell/\lambda$ and $v = u/\lambda$,

$$F(v) = \int_{-\infty}^{\infty} d\xi [1 - \mathbf{n}_0(\xi - v/2) \cdot \mathbf{n}_0(\xi + v/2)]$$

$$= (\cosh v - 1) \int_{-\infty}^{\infty} \frac{d\xi}{\cosh(\xi - v/2) \cosh(\xi + v/2)}$$

$$= 2v \tanh(v/2)$$

$$= v^2 - \frac{v^4}{12} + O(v^6), \quad F(v) \sim 2|v| \quad (|v| \gg 1). \quad (\text{S46})$$

At $\delta R = 0$ in AM, use $v(\varphi) = 2\bar{R}e_s \cos 2\varphi/\lambda$. At fixed e_s and λ ,

$$\frac{\partial E_{\text{inter}}}{\partial \bar{R}} \simeq A_0 t \lambda \int_0^{2\pi} [F(v) + vF'(v)] d\varphi,$$

$$F'(v) = 2 \tanh(v/2) + v \operatorname{sech}^2(v/2).$$

For the antiferromagnetic coupling $A_0 > 0$, $F(v) \geq 0$ and $vF'(v) \geq 0$. Thus the local estimate increases with $\bar{R}$ when $e_s \neq 0$. In the zero-field PAR state, $u = 0$ and $F(0) = 0$, so $E_{\text{inter}} = 0$.

For an in-plane unit normal $\hat{\mathbf{n}} = (n_x, n_y)$, the single-sublattice wall-width estimates are

$$\lambda_A(\hat{\mathbf{n}}) = \sqrt{\frac{A_1 n_x^2 + A_2 n_y^2}{K}},$$

$$\lambda_B(\hat{\mathbf{n}}) = \sqrt{\frac{A_2 n_x^2 + A_1 n_y^2}{K}}.$$

Along x ,

$$\lambda_A = \lambda \rho^{1/4}, \quad \lambda_B = \lambda \rho^{-1/4},$$

$$\frac{\lambda_A}{\lambda_B} = \sqrt{\rho} = 2 \quad (\rho = 4).$$

The corresponding estimates are equal between sublattices in PAR. The common- λ calculation above therefore omits the difference between the two directional wall-width scales. The numerical states minimize the full energy without imposing a common wall width.

S10. ENERGY OF STATES MINIMIZED AT FIXED q

For AM at $\rho = 2$ and $B_z = 0$, the constrained coordinate is

$$q = \sqrt{S_A/\pi} - \sqrt{S_B/\pi},$$

$$S_\eta = \frac{h^2}{2} \sum_{ij} (1 + s_\eta m_{\eta z, ij}), \quad (\text{S47})$$

$$s_A = +1, \quad s_B = -1.$$

For prescribed q^* , we minimize

$$E_{\text{aug}} = E + \Lambda_q(q - q^*) + \frac{k_p}{2}(q - q^*)^2,$$

where $[\Lambda_q] = \text{J/m}$ and $[k_p] = \text{J/m}^2$. The constraint gradient is obtained from

$$\frac{\partial q}{\partial \mathbf{m}_{\eta, ij}} = \frac{h^2}{4\sqrt{\pi}S_\eta} \hat{\mathbf{e}}_z, \quad \eta = A, B,$$

$$\mathbf{g}_{\eta, ij}^{\text{aug}} = -\frac{1}{h^2 t} \left[\frac{\partial E}{\partial \mathbf{m}_{\eta, ij}} + [\Lambda_q + k_p(q - q^*)] \frac{\partial q}{\partial \mathbf{m}_{\eta, ij}} \right].$$

The tangent projection and norms are defined as in Sec. S2, now using $\mathbf{g}^{\text{aug}}$. Accepted states satisfy

$$g_{\text{max}}^{\text{aug}} \leq 1 \text{ J/m}^3,$$

$$g_{\text{rms}}^{\text{aug}} \leq 0.1 \text{ J/m}^3,$$

$$|q - q^*| \leq 0.002 \text{ nm}.$$

When the constraint tolerance is not met, the multiplier is updated by

$$\Lambda_q \leftarrow \Lambda_q + k_p(q - q^*).$$

From the fourth outer iteration onward, the update also includes $k_p \leftarrow 1.5k_p$. After minimization, we record the physical energy E and measure $\delta R = \sqrt{a_A b_A} - \sqrt{a_B b_B}$ from the contours. The constraint coordinate q includes the wall magnetization through S_η and is not set equal to δR .

The accepted targets are

$$q_j^* = 0.25j \text{ nm}, \quad j = -12, \dots, 12.$$

These 25 states retain both cores and give $|\delta R| \lesssim 2.8046 \text{ nm}$. At $q^* = \pm 3.25 \text{ nm}$ one core disappears. For the states with $|\delta R_j| \leq 0.8 \text{ nm}$, the least-squares fit is

$$(E_c, b_R, \kappa_{\text{con}})$$

$$= \operatorname{argmin}_{c, b, k} \sum_j \left[E_j - c - b \delta R_j - \frac{k}{2} (\delta R_j)^2 \right]^2.$$

It gives

$$\kappa_{\text{con}} = 12.1658 \text{ mJ/m}^2.$$

The intercept E_c and linear coefficient b_R are fitted independently. The reported curvature is twice the quadratic polynomial coefficient. This coefficient describes the energy along states minimized at fixed q , plotted against their measured δR . Section S15 obtains P from the same states.

S11. LLG INTEGRATION, HELICITY, AND OSCILLATION FITS

Let $\mathbf{r}_{\eta, j}^{(0)}$ be the 181 sampling positions on the zero-field contour,

$$\mathbf{r}_{\eta, j}^{(0)} = r_\eta^{(0)}(\varphi_j)(\cos \varphi_j, \sin \varphi_j).$$

The positions remain fixed during the LLG calculation. The spin components at these positions are interpolated bilinearly. Define

$$Z_\eta(t) = \sum_j \left[m_{\eta x}(\mathbf{r}_{\eta, j}^{(0)}, t) + i m_{\eta y}(\mathbf{r}_{\eta, j}^{(0)}, t) \right] e^{-i\varphi_j}.$$

This is identical to the stated $|\mathbf{m}_{\eta\parallel}|$ -weighted sum because

$$|\mathbf{m}_{\eta\parallel}| e^{i \operatorname{atan2}(m_{\eta y}, m_{\eta x})} = m_{\eta x} + i m_{\eta y}.$$

The helicity deviation is

$$\phi_\eta(t) = \text{unwrap}[\arg Z_\eta(t)] - \phi_\eta^{(0)}, \quad \phi_\eta^{(0)} = \arg Z_\eta^{(0)},$$

where $Z_\eta^{(0)}$ is evaluated in the zero-field equilibrium state.

The main-text angles are

$$\bar{\phi}(t) = \frac{\phi_A(t) + \phi_B(t)}{2}, \quad \phi_-(t) = \frac{\phi_A(t) - \phi_B(t)}{2}.$$

For a common spin rotation by β ,

$$Z_\eta \mapsto e^{i\beta} Z_\eta, \quad \phi_\eta \mapsto \phi_\eta + \beta, \quad \bar{\phi} \mapsto \bar{\phi} + \beta.$$

The pulse ends at $t = 0$:

$$B_z(t) = \begin{cases} 0.05 \text{ T}, & -10 \text{ ps} \leq t < 0, \\ 0, & 0 \leq t \leq 2 \text{ ns}. \end{cases}$$

We integrate main-text Eq. (3) with $\alpha = 10^{-3}$ and $\Delta t_{\text{LLG}} = 4 \text{ fs}$. Writing its right-hand side for the full spin array as $\mathbf{v}[\mathbf{m}, t]$, the fourth-order Runge–Kutta step is

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{v}[\mathbf{m}^\ell, t_\ell^{\text{LLG}}], \\ \mathbf{k}_2 &= \mathbf{v}[\mathbf{m}^\ell + \frac{1}{2}\Delta t_{\text{LLG}}\mathbf{k}_1, t_\ell^{\text{LLG}} + \frac{1}{2}\Delta t_{\text{LLG}}], \\ \mathbf{k}_3 &= \mathbf{v}[\mathbf{m}^\ell + \frac{1}{2}\Delta t_{\text{LLG}}\mathbf{k}_2, t_\ell^{\text{LLG}} + \frac{1}{2}\Delta t_{\text{LLG}}], \\ \mathbf{k}_4 &= \mathbf{v}[\mathbf{m}^\ell + \Delta t_{\text{LLG}}\mathbf{k}_3, t_\ell^{\text{LLG}} + \Delta t_{\text{LLG}}], \\ \mathbf{m}_{\text{pre}}^{\ell+1} &= \mathbf{m}^\ell + \frac{\Delta t_{\text{LLG}}}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \end{aligned}$$

The complete effective field, including magnetostatics, is recomputed in every evaluation of $\mathbf{v}$. Each sublattice spin is then normalized separately:

$$\mathbf{m}_{\eta,ij}^{\ell+1} = \frac{\mathbf{m}_{\eta,ij,\text{pre}}^{\ell+1}}{|\mathbf{m}_{\eta,ij,\text{pre}}^{\ell+1}|}.$$

The seven values of ρ in both AM and PAR give 14 LLG calculations in the 96 nm periodic cell. The saved series contains

$$\begin{aligned} N_t &= 4001, & \Delta t_{\text{save}} &= 0.5 \text{ ps}, \\ t_n &= n\Delta t_{\text{save}}, & n &= 0, \dots, N_t - 1. \end{aligned}$$

For a coordinate X to which the damped-cosine fit is applied, the model is

$$\begin{aligned} X_{\text{fit}}(t) &= c_X + A_X e^{-t/\tau_X} \cos(2\pi f_X t + \psi_X), \\ 0 &\leq t \leq 2 \text{ ns}. \end{aligned}$$

Here $|A_X|$ is the envelope amplitude at $t = 0$, τ_X is the decay time, f_X is the frequency, ψ_X is the phase at $t = 0$, and c_X is a constant offset. The Fourier frequencies and amplitudes are defined below.

For AM at $\rho = 2$, the time-domain fits give

$$f_{\delta R} = 49.52188 \text{ GHz}, \quad \psi_{\bar{\phi}} - \psi_{\delta R} = 90.01048^\circ.$$

For each sampled coordinate $X_n = X(t_n)$, define

$$\langle X \rangle_t = \frac{1}{N_t} \sum_{n=0}^{N_t-1} X_n, \quad f_k = \frac{k}{N_t \Delta t_{\text{save}}},$$

$$\hat{X}(f_k) = \frac{2}{\sum_{n=0}^{N_t-1} w_n} \sum_{n=0}^{N_t-1} w_n [X_n - \langle X \rangle_t] e^{-2\pi i k n / N_t},$$

$$A_X^{\text{F}}(f_k) = |\hat{X}(f_k)|.$$

The symmetric Hann weights are

$$w_n = \frac{1}{2} \left[1 - \cos \left(\frac{2\pi n}{N_t - 1} \right) \right], \quad n = 0, \dots, N_t - 1.$$

The transform uses N_t points without zero padding. The initial peak search includes every nonzero positive frequency:

$$k_{\text{init}} = \underset{k \in \mathcal{K}}{\text{argmax}} A_{\delta R}^{\text{F}}(f_k), \quad \mathcal{K} = \{1, \dots, (N_t - 1)/2\}.$$

After the time-domain fit, reported Fourier amplitudes use

$$k_* = \underset{k \in \mathcal{K}}{\text{argmin}} |f_k - f_{\delta R}|.$$

For the $\rho = 2$ and 4 AM data reported here, $k_* = k_{\text{init}}$. The positive-frequency spacing is

$$\Delta f = \frac{1}{N_t \Delta t_{\text{save}}} = 0.499875 \dots \text{ GHz}.$$

For the selected Fourier frequency at $\rho = 2$,

$$k_* = 99, \quad f_{k_*} = 49.4876 \text{ GHz}.$$

The amplitudes evaluated at this same frequency are

X	$A_X^{\text{F}}(f_{k_*})$	unit
δR	0.2068795	nm
$\bar{\phi}$	0.03257398	rad
ϕ_-	5.956134×10^{-8}	rad
e_u	1.445541×10^{-5}	1
$\bar{R}$	2.354173×10^{-7}	nm

Therefore,

$$\begin{aligned} \frac{A_{\phi_-}^{\text{F}}(f_{k_*})}{A_{\bar{\phi}}^{\text{F}}(f_{k_*})} &= 1.82849 \times 10^{-6}, \\ \frac{A_R^{\text{F}}(f_{k_*})}{A_{\delta R}^{\text{F}}(f_{k_*})} &= 1.13794 \times 10^{-6}, \\ \frac{A_{e_u}^{\text{F}}(f_{k_*})}{A_{\delta R}^{\text{F}}(f_{k_*})} &= 6.98736 \times 10^{-5} \text{ nm}^{-1}. \end{aligned}$$

These comparisons concern f_{k_*} .

The projections of $\bar{\phi}$ and e_u fix the frequency and decay time to those obtained from $\delta R(t)$, with a free offset and cosine and sine coefficients for each coordinate. The phase of δR is represented with a positive amplitude. For the e_u projection, define

$$\begin{aligned}\theta_{\text{osc}}(t) &= 2\pi f_{\delta R} t + \psi_{\delta R}, \\ C(t) &= e^{-t/\tau_{\delta R}} \cos \theta_{\text{osc}}(t), \\ S(t) &= e^{-t/\tau_{\delta R}} \sin \theta_{\text{osc}}(t).\end{aligned}$$

The projected series is

$$e_u^{\text{proj}}(t) = c_e + a_e C(t) + b_e S(t).$$

The unweighted least-squares projection includes a free constant:

$$(c_e, a_e, b_e) = \underset{c, a, b}{\text{argmin}} \sum_n [e_u(t_n) - c - aC(t_n) - bS(t_n)]^2.$$

Its amplitude and phase relative to δR are

$$A_e^{\text{proj}} = \sqrt{a_e^2 + b_e^2}, \quad \Delta\psi_e^{\text{proj}} = \text{atan2}(-b_e, a_e).$$

The reported values are

$$\begin{aligned}\frac{A_e^{\text{proj}}}{|A_{\delta R}|} &= 1.236 \times 10^{-4} \text{ nm}^{-1}, \\ \Delta\psi_e^{\text{proj}} &= -0.653^\circ, \quad \tau_{\delta R} = 3.2165 \text{ ns}.\end{aligned}$$

The quoted decay time is inherited from the δR fit.

For each value of ρ , let k_* denote the frequency index selected as above. The surrounding ϕ_- amplitude is defined by

$$\begin{aligned}A_{\phi_-, \text{bg}}^{\text{F}} &= \text{median}_{j \in \mathcal{J}_{\text{bg}}} A_{\phi_-}^{\text{F}}(f_{k_*+j}), \\ \mathcal{J}_{\text{bg}} &= \{-5, -4, -3, -2, 2, 3, 4, 5\}.\end{aligned}$$

For AM at $\rho = 4$,

$$\frac{A_{\phi_-}^{\text{F}}(f_{k_*})}{A_{\phi_-}^{\text{F}}(f_{k_*})} = 1.53392 \times 10^{-5}, \quad \frac{A_{\phi_-}^{\text{F}}(f_{k_*})}{A_{\phi_-, \text{bg}}^{\text{F}}} = 23.84.$$

Section S15 compares the oscillation with the prediction from static coefficients.

S12. BERRY COUPLING OF RADIUS DIFFERENCE AND HELICITY

Let $\mathbf{m}_\eta^{(q)}(\mathbf{r})$ be the zero-field states minimized at prescribed q in Sec. S10. Locally near $q = 0$, the trial motion is

$$\begin{aligned}\mathbf{m}_\eta(\mathbf{r}, t') &= R_z[\bar{\phi}(t')] \mathbf{m}_\eta^{(q(t'))}(\mathbf{r}), \\ \delta R(0) &= 0, \quad \left. \frac{d\delta R}{dq} \right|_0 \neq 0.\end{aligned}$$

Here R_z rotates the spins, and t' is the time variable. The prefactor $M_s t$ below contains the film thickness. We assume a smooth local continuation of this family, so either q or its measured δR can parametrize the radial motion.

Write the local unit spin as

$$\mathbf{m}_\eta = (\sin \theta_\eta \cos \Phi_\eta, \sin \theta_\eta \sin \Phi_\eta, \cos \theta_\eta).$$

The Berry action is

$$S_B = \frac{M_s t}{\gamma} \sum_{\eta=A,B} \int dt' \int d^2 r \Omega_\eta(\theta_\eta) \dot{\Phi}_\eta. \quad (\text{S48})$$

The overdot denotes differentiation with respect to time. We choose

$$\begin{aligned}\Omega_\eta &= -\cos \theta_\eta - s_\eta, \\ \Omega_A &= -(1 + \cos \theta_A), \quad \Omega_B = 1 - \cos \theta_B.\end{aligned} \quad (\text{S49})$$

Then $\partial_{\theta_\eta} \Omega_\eta = \sin \theta_\eta$, with the precession sign of main-text Eq. (3), and $\Omega_\eta \rightarrow 0$ in the backgrounds $\cos \theta_\eta \rightarrow -s_\eta$. Define

$$I_A = \int d^2 r (1 + \cos \theta_A), \quad I_B = \int d^2 r (1 - \cos \theta_B). \quad (\text{S50})$$

These integrands vanish in the corresponding backgrounds and satisfy $I_\eta = 2S_\eta$.

For uniform spin-angle increments,

$$\Phi_\eta(\mathbf{r}, t') = \Phi_\eta^{(q(t'))}(\mathbf{r}) + \phi_\eta(t'), \quad \dot{\Phi}_\eta = \partial_q \Phi_\eta^{(q)} \dot{q} + \dot{\phi}_\eta.$$

The term proportional to $\dot{q}$ is $\mathcal{A}(q)\dot{q}$, where

$$\mathcal{A}(q) = \frac{M_s t}{\gamma} \sum_\eta \int d^2 r \Omega_\eta(\theta_\eta^{(q)}) \partial_q \Phi_\eta^{(q)}.$$

Locally along this one-parameter family,

$$\mathcal{A}(q)\dot{q} = \frac{d}{dt'} \left[\int_0^{q(t')} \mathcal{A}(q') dq' \right].$$

After removing this total time derivative, the Berry action is

$$S_B = \frac{M_s t}{\gamma} \int dt' [-I_A \dot{\phi}_A + I_B \dot{\phi}_B]. \quad (\text{S51})$$

Using $\phi_A = \bar{\phi} + \phi_-$ and $\phi_B = \bar{\phi} - \phi_-$ gives

$$S_B = -\frac{M_s t}{\gamma} \int dt' [(I_A - I_B) \dot{\bar{\phi}} + (I_A + I_B) \dot{\phi}_-]. \quad (\text{S52})$$

For these uniform increments about the zero-field helicities,

$$\mathcal{P} : (\phi_A, \phi_B) \mapsto (-\phi_B, -\phi_A), \quad (\bar{\phi}, \phi_-) \mapsto (-\bar{\phi}, \phi_-).$$

The exact identity

$$M_s t (I_A - I_B) = M_s t \int d^2 r (m_{Az} + m_{Bz}) = \mu_z$$

connects the common spin rotation to the net moment. For the locally $\mathcal{P}$ -related states at q and $-q$, μ_z and δR are odd. Thus

$$I_A - I_B = \left. \frac{d(I_A - I_B)}{d\delta R} \right|_0 \delta R + O(\delta R^3). \quad (\text{S53})$$

The common-rotation trial has $\phi_- = 0$. Its leading Berry term is

$$L_B = -P\delta R\dot{\phi}, \quad P = \frac{M_s t}{\gamma} \left. \frac{d(I_A - I_B)}{d\delta R} \right|_0 = \frac{1}{\gamma} \left. \frac{d\mu_z}{d\delta R} \right|_0.$$

Both derivatives are taken along the same fixed- q family. Equivalently,

$$P = \left. \frac{1}{\gamma} \frac{d\mu_z/dq}{d\delta R/dq} \right|_{q=0}.$$

The quadratic term in the radial energy is $\frac{1}{2}\kappa_{\text{con}}\delta R(q)^2$, with κ_{con} obtained from the polynomial fit in Sec. S10. The common spin rotation leaves $m_{\eta z}$ and therefore the measured $\delta R, \bar{R}, e_s, e_u$ unchanged. In particular, $e_u(t') = e_u[\mathbf{m}^{(q(t'))}]$ within this trial family. The radial energy and Berry term enter the Lagrangian in Sec. S14.

S13. ENERGY OF A COMMON SPIN ROTATION

For the zero-field state at $\rho = 2$, rotate each spin by the same angle while keeping its position fixed:

$$\mathbf{m}_\eta(\mathbf{r}; \bar{\phi}) = R_z(\bar{\phi})\mathbf{m}_\eta^{(0)}(\mathbf{r}),$$

$$R_z(\bar{\phi}) = \begin{pmatrix} \cos \bar{\phi} & -\sin \bar{\phi} & 0 \\ \sin \bar{\phi} & \cos \bar{\phi} & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The exchange, anisotropy, and intersublattice terms are unchanged. The separate second derivatives are

$$\kappa_{\bar{\phi}}^{\text{DMI}} = \left. \frac{d^2 E_D[R_z(\bar{\phi})\mathbf{m}^{(0)}]}{d\bar{\phi}^2} \right|_0,$$

$$\kappa_{\bar{\phi}}^{\text{ms}} = \left. \frac{d^2 E_{\text{ms}}[R_z(\bar{\phi})\mathbf{m}^{(0)}]}{d\bar{\phi}^2} \right|_0.$$

The $\mathcal{Q}$ symmetry in Sec. S14 makes the energy even in $\bar{\phi}$, so

$$E(\bar{\phi}) - E(0) = \frac{1}{2}\kappa_{\bar{\phi}}\bar{\phi}^2 + O(\bar{\phi}^4).$$

Their sum and the circular-wall estimate are

$$\kappa_{\bar{\phi}} = \kappa_{\bar{\phi}}^{\text{DMI}} + \kappa_{\bar{\phi}}^{\text{ms}}, \quad \kappa_{\bar{\phi}}^{\text{DMI}} \simeq 4\pi^2 D \bar{R} t. \quad (\text{S54})$$

The final estimate is derived below for circular walls with $\lambda/R_\eta \ll 1$. The numerical coefficient is obtained by rotating the complete calculated state.

For a circular Néel texture, write

$$\Phi_\eta = \varphi + \phi_\eta^{(0)} + \bar{\phi}, \quad \int_0^\infty \theta'_\eta(r) dr = s_\eta \pi,$$

with core and background polarities $m_{\eta z}(0) = s_\eta$ and $m_{\eta z}(\infty) = -s_\eta$. For the chosen $D > 0$, the lower-energy Néel orientations satisfy $\cos \phi_\eta^{(0)} = -s_\eta$. The DMI energy of each sublattice is

$$E_{D,\eta}(\bar{\phi}) = 2\pi D t \cos(\phi_\eta^{(0)} + \bar{\phi}) \times \int_0^\infty [r\theta'_\eta(r) + \sin \theta_\eta \cos \theta_\eta] dr.$$

For a smooth wall of width $\lambda \ll R_\eta$,

$$\int_0^\infty r\theta'_\eta(r) dr \simeq s_\eta \pi R_\eta,$$

$$\left| \int_0^\infty \sin \theta_\eta \cos \theta_\eta dr \right| = O(\lambda).$$

Keeping the leading term gives

$$E_{D,\eta}(\bar{\phi}) \simeq -2\pi^2 D t R_\eta \cos \bar{\phi},$$

$$E_D(\bar{\phi}) \simeq -4\pi^2 D t \bar{R} \cos \bar{\phi},$$

$$\left. \frac{d^2 E_D}{d\bar{\phi}^2} \right|_0 \simeq 4\pi^2 D t \bar{R}.$$

The numerical rotations use

$$\bar{\phi} \in \{0, \pm 0.01, \pm 0.02, \pm 0.04, \pm 0.08\} \text{ rad}.$$

For each contribution $a \in \{D, \text{ms}\}$, the unweighted fit is

$$(E_{a,\text{off}}, b_a, k_a) = \underset{c,b,k}{\text{argmin}} \sum_{|\bar{\phi}_j| \leq 0.04} \left[E_a(\bar{\phi}_j) - c - b\bar{\phi}_j - \frac{k}{2}\bar{\phi}_j^2 \right]^2.$$

The coefficients k_D and k_{ms} estimate $\kappa_{\bar{\phi}}^{\text{DMI}}$ and $\kappa_{\bar{\phi}}^{\text{ms}}$.

Quadratic fits using $|\bar{\phi}| \leq 0.04$ rad give

$$\kappa_{\bar{\phi}}^{\text{DMI}} = 4.09917880 \times 10^{-19} \text{ J},$$

$$\kappa_{\bar{\phi}}^{\text{ms}} = -1.43448804 \times 10^{-21} \text{ J}, \quad (\text{S55})$$

$$\kappa_{\bar{\phi}} = 4.08483392 \times 10^{-19} \text{ J}.$$

Varying the maximum included angle from 0.02 to 0.08 rad gives the range

$$\kappa_{\bar{\phi}} \in [4.08317, 4.08526] \times 10^{-19} \text{ J},$$

$$\frac{4.08526 - 4.08317}{4.08483392} \simeq 0.051\%.$$

For a magnetization source $\mathbf{U}$, denote the quadratic magnetostatic energy of Sec. S1 by $E_{\text{ms}}[\mathbf{U}]$. Under the common rotation,

$$\mathbf{M}_\parallel(\bar{\phi}) = \cos \bar{\phi} \mathbf{M}_\parallel^{(0)} + \sin \bar{\phi} (\hat{\mathbf{e}}_z \times \mathbf{M}_\parallel^{(0)}),$$

$$M_z(\bar{\phi}) = M_z^{(0)}.$$

Using the quadratic energy and $N_{xz} = N_{yz} = 0$ gives

$$\kappa_{\bar{\phi}}^{\text{ms}} = 2 \left[E_{\text{ms}}[\hat{\mathbf{e}}_z \times \mathbf{M}_{\parallel}^{(0)}] - E_{\text{ms}}[\mathbf{M}_{\parallel}^{(0)}] \right].$$

Each energy on the right is nonnegative, but their difference can have either sign. For the calculated state,

$$\frac{\kappa_{\bar{\phi}}^{\text{ms}}}{\kappa_{\bar{\phi}}^{\text{DMI}}} = -0.350\%, \quad \kappa_{\bar{\phi}} > 0.$$

S14. SYMMETRY AND EQUATIONS OF THE TWO-COORDINATE MODEL

Under $\mathcal{P}$, $(\delta R, \bar{\phi}) \mapsto (-\delta R, -\bar{\phi})$, so $\delta R \bar{\phi}$ is unchanged. To test this potential-energy term, use the static transformation

$$\mathcal{Q}: \quad \mathbf{m}_{\eta}(x, y) \mapsto (-m_{\eta x}, m_{\eta y}, m_{\eta z})(-x, y). \quad (\text{S56})$$

This is reflection in x combined with time reversal. Let $R_x = \text{diag}(-1, 1, 1)$. The local energy densities satisfy

$$w_a[\mathcal{Q}\mathbf{m}](x, y) = w_a[\mathbf{m}](-x, y), \\ a \in \{\text{ex}, K, D, \text{inter}, Z\}.$$

with B_z unchanged. For magnetostatics,

$$\widehat{\mathbf{M}}'(\mathbf{k}) = R_x \widehat{\mathbf{M}}(R_x \mathbf{k}), \quad N(R_x \mathbf{k}) = R_x N(\mathbf{k}) R_x^T,$$

so $E_{\text{ms}}[\mathcal{Q}\mathbf{m}] = E_{\text{ms}}[\mathbf{m}]$. The contour and spin angles transform as

$$r_{\eta}(\varphi) \mapsto r_{\eta}(\pi - \varphi), \\ (\bar{R}, \delta R, e_s, e_u) \mapsto (\bar{R}, \delta R, e_s, e_u), \\ (\bar{\phi}, \phi_-) \mapsto (-\bar{\phi}, -\phi_-).$$

For the selected zero-field family preserving $\mathcal{Q}$,

$$E(\delta R, \bar{\phi}) = E(\delta R, -\bar{\phi}), \quad \left. \frac{\partial^2 E}{\partial \delta R \partial \bar{\phi}} \right|_0 = 0.$$

Thus the quadratic potential contains no $\delta R \bar{\phi}$ term.

Without damping, the quadratic Lagrangian for this trial family is

$$L_{\text{var}} = -P \delta R \dot{\bar{\phi}} - \frac{1}{2} \kappa_{\text{con}} \delta R^2 - \frac{1}{2} \kappa_{\bar{\phi}} \bar{\phi}^2. \quad (\text{S57})$$

The Euler–Lagrange equations give

$$P \delta \dot{R} = \kappa_{\bar{\phi}} \bar{\phi}, \quad P \dot{\bar{\phi}} = -\kappa_{\text{con}} \delta R.$$

Eliminating $\bar{\phi}$ gives

$$\frac{P^2}{\kappa_{\bar{\phi}}} \delta \ddot{R} = -\kappa_{\text{con}} \delta R, \quad m^* = \frac{P^2}{\kappa_{\bar{\phi}}}. \quad (\text{S58})$$

Here m^* is the mass associated with δR , with units $\text{J s}^2/\text{m}^2 = \text{kg}$. For $\kappa_{\text{con}} > 0$, $\kappa_{\bar{\phi}} > 0$, and $P \neq 0$,

$$\omega_0 = \frac{\sqrt{\kappa_{\text{con}} \kappa_{\bar{\phi}}}}{|P|}, \quad f_0 = \frac{\omega_0}{2\pi}.$$

Choosing $\delta R(t') = A_{\delta R} \cos(\omega_0 t' + \psi_{\delta R})$ gives

$$\bar{\phi}(t') = -\frac{P \omega_0}{\kappa_{\bar{\phi}}} A_{\delta R} \sin(\omega_0 t' + \psi_{\delta R}).$$

Therefore the undamped model predicts

$$\frac{|\bar{\phi}|}{|\delta R|} = \sqrt{\frac{\kappa_{\text{con}}}{\kappa_{\bar{\phi}}}}, \quad \psi_{\bar{\phi}} - \psi_{\delta R} = \text{sgn}(P) \frac{\pi}{2}.$$

Internal spin deformations can give skyrmion inertia [3, 4]. Combining the piecewise m_z integrals of Sec. S6 with the circular-wall DMI estimate of Sec. S13 gives

$$I_{\eta} \simeq 2\pi R_{\eta}^2, \quad I_A - I_B \simeq 4\pi \bar{R} \delta R, \quad P \simeq \frac{4\pi M_s t \bar{R}}{\gamma}.$$

If $\lambda/\bar{R} \ll 1$ and the magnetostatic rotation term is omitted,

$$\kappa_{\bar{\phi}} \simeq 4\pi^2 D t \bar{R}, \quad m^* \simeq \frac{4M_s^2 t \bar{R}}{\gamma^2 D}.$$

For the calculated reference state, $\lambda/\bar{R} \simeq 4.85/3.87 \simeq 1.25$. The numerical prediction in Sec. S15 uses P and $\kappa_{\bar{\phi}}$ evaluated from the complete spin configurations.

S15. COMPARISON WITH THE LLG OSCILLATION

For AM at $\rho = 2$, use the seven fixed- q states with measured $|\delta R_j| \leq 0.8 \text{ nm}$, selected as $|q^*|$ increases from zero. The unweighted fit is

$$(I_{\text{off}}, s_I) = \underset{c, s}{\text{argmin}} \sum_{j=1}^7 [(I_{A,j} - I_{B,j}) - c - s \delta R_j]^2.$$

The intercept I_{off} is independent. The slope s_I has units of length and estimates the derivative at zero:

$$\left. \frac{d(I_A - I_B)}{d\delta R} \right|_0 \simeq s_I = 6.42339469 \times 10^{-8} \text{ m},$$

$$P \simeq \frac{M_s t}{\gamma} s_I = 2.11680052 \times 10^{-22} \text{ J s/m}. \quad (\text{S59})$$

The static inputs at $\rho = 2$ are

$$\kappa_{\text{con}} = 0.01216576558 \text{ J/m}^2, \\ \kappa_{\bar{\phi}} = 4.08483392 \times 10^{-19} \text{ J}.$$

together with P above. The first two are obtained from the fixed- q energy fit and the common spin rotation, respectively. The two-coordinate model then gives

$$f_0 = \frac{\sqrt{\kappa_{\text{con}}\kappa_{\bar{\phi}}}}{2\pi|P|} = 53.00259 \text{ GHz},$$

$$\left(\frac{|\bar{\phi}|}{|\delta R|}\right)_{\text{model}} = \sqrt{\frac{\kappa_{\text{con}}}{\kappa_{\bar{\phi}}}} = 1.72576843 \times 10^8 \text{ rad/m.} \quad (\text{S60})$$

The LLG time-domain fits of Sec. S11 give

$$f_{\text{LLG}} = 49.52188 \text{ GHz},$$

$$\frac{|A_{\bar{\phi}}|}{|A_{\delta R}|} = 1.57493011 \times 10^8 \text{ rad/m.}$$

$$\psi_{\bar{\phi}} - \psi_{\delta R} = 90.01048^\circ.$$

The model phase is $+90^\circ$ for the positive P above. Using the unrounded static inputs, the relative differences are

$$\left(\frac{f_0}{f_{\text{LLG}}} - 1\right) = 7.029\%,$$

$$\left[\frac{\sqrt{\kappa_{\text{con}}/\kappa_{\bar{\phi}}}}{|A_{\bar{\phi}}|/|A_{\delta R}|} - 1\right] = 9.577\%.$$

The same inputs give

$$m^* = P^2/\kappa_{\bar{\phi}} = 1.09694654 \times 10^{-25} \text{ kg.}$$

All three static inputs were extracted at $\rho = 2$ only. The equality $Z_\eta \mapsto e^{i\beta}Z_\eta$ in Sec. S11 shows that a uniform spin rotation changes the measured helicity by the same angle, connecting the model coordinate to the LLG observable.

On the same sequence of fixed- q states, define

$$\mathcal{J}_X = \{j : |\delta R_j| \leq X\}, \quad e_{u,j} \simeq c_e(X) + s_e(X)\delta R_j.$$

The linear fits give

X (nm)	$ \mathcal{J}_X $	$s_e(X)$ (nm $^{-1}$)
0.40	3	-7.58×10^{-4}
0.60	5	$+3.39 \times 10^{-4}$
0.80	7	$+5.94 \times 10^{-4}$
1.00	9	$+5.34 \times 10^{-4}$

The nearest nonzero measured radius differences satisfy

$$\min_{j:\delta R_j \neq 0} |\delta R_j| \simeq 0.2392 \text{ nm.}$$

For comparison, the quadratic static relation from main-text Eq. (9), using the interval-based estimate in Sec. S7, would give

$$e_u = -\frac{\zeta}{\kappa_{e_u}}\delta R, \quad -\frac{\zeta}{\kappa_{e_u}} \simeq -3.49 \times 10^{-3} \text{ nm}^{-1}.$$

The LLG projection defined in Sec. S11 instead gives

$$\frac{A_e^{\text{proj}}}{|A_{\delta R}|} = 1.236 \times 10^{-4} \text{ nm}^{-1}, \quad \Delta\psi_e^{\text{proj}} = -0.653^\circ.$$

Its Fourier amplitude ratio at f_{k_*} is separately

$$\frac{A_{e_u}^{\text{F}}(f_{k_*})}{A_{\delta R}^{\text{F}}(f_{k_*})} = 6.98736 \times 10^{-5} \text{ nm}^{-1}.$$

Within the variational model,

$$e_u^{\text{trial}}(t) = e_u[\mathbf{m}^{(q(t))}],$$

and P and κ_{con} are obtained from this same fixed- q family, as specified in Sec. S12.

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